

BFS Traversal

Step 1: Start at E

Legend

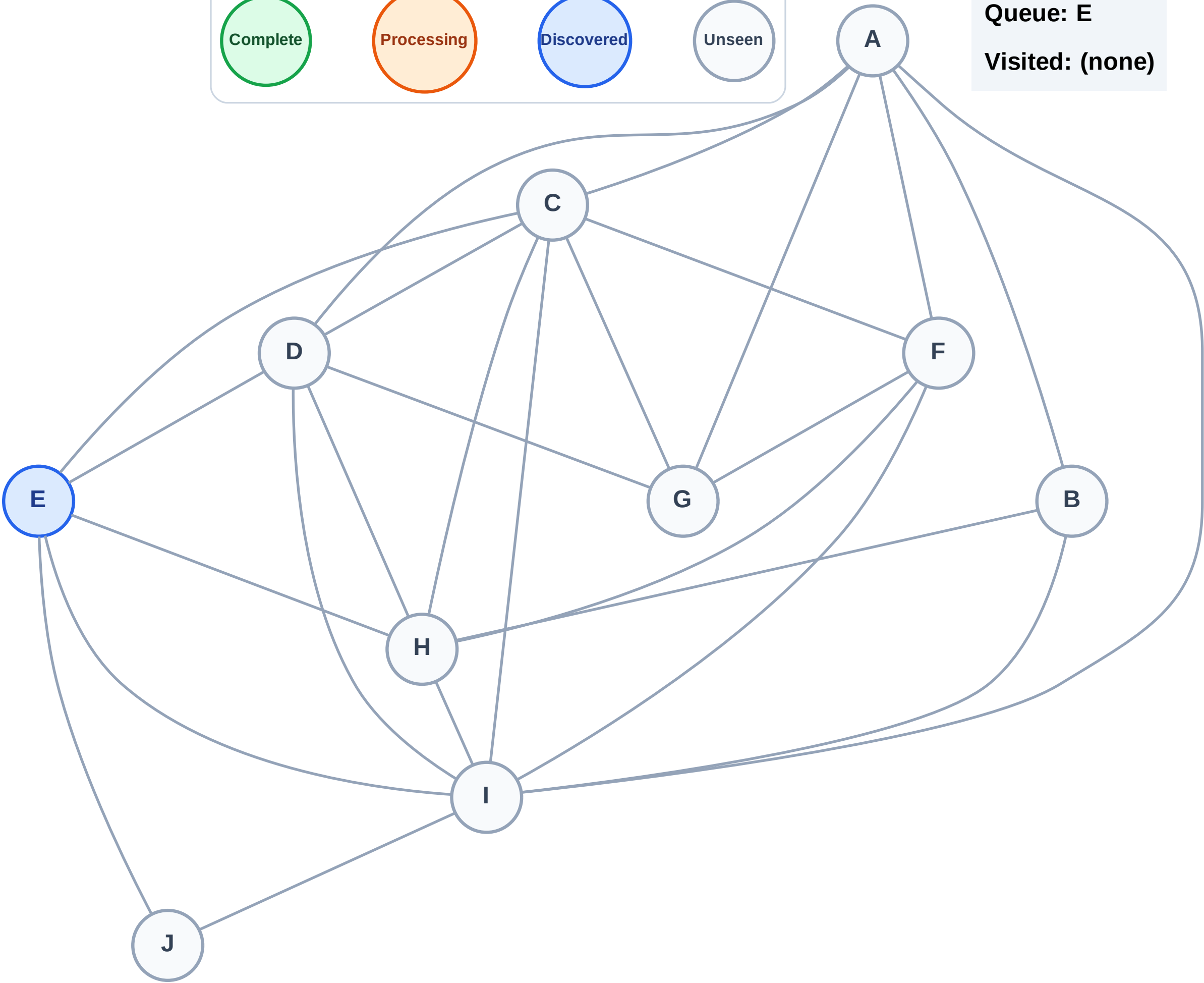
Complete

Processing

Discovered

Unseen

Queue: E
Visited: (none)



BFS Traversal

Step 2: Process node E

Legend

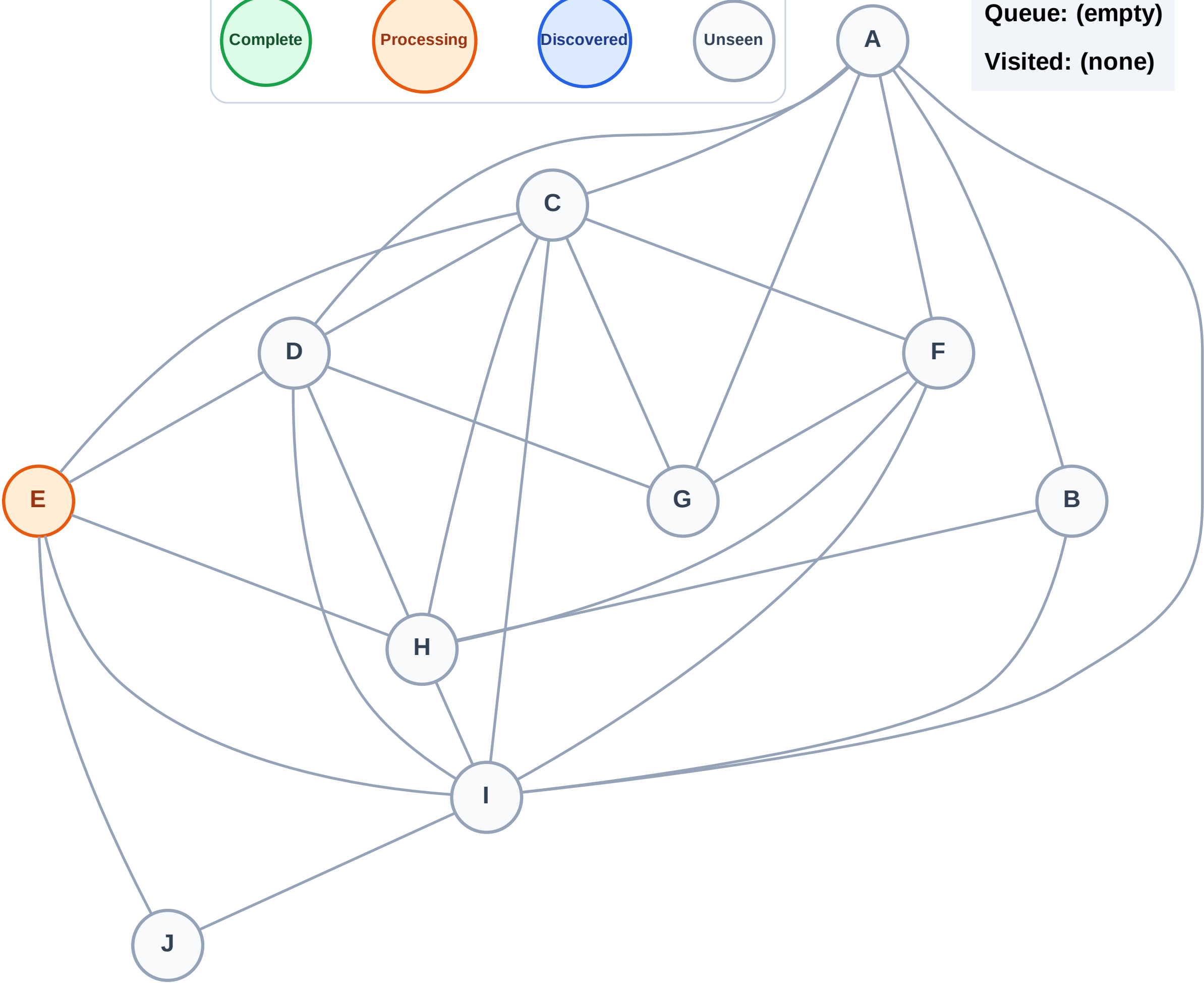
Complete

Processing

Discovered

Unseen

Queue: (empty)
Visited: (none)



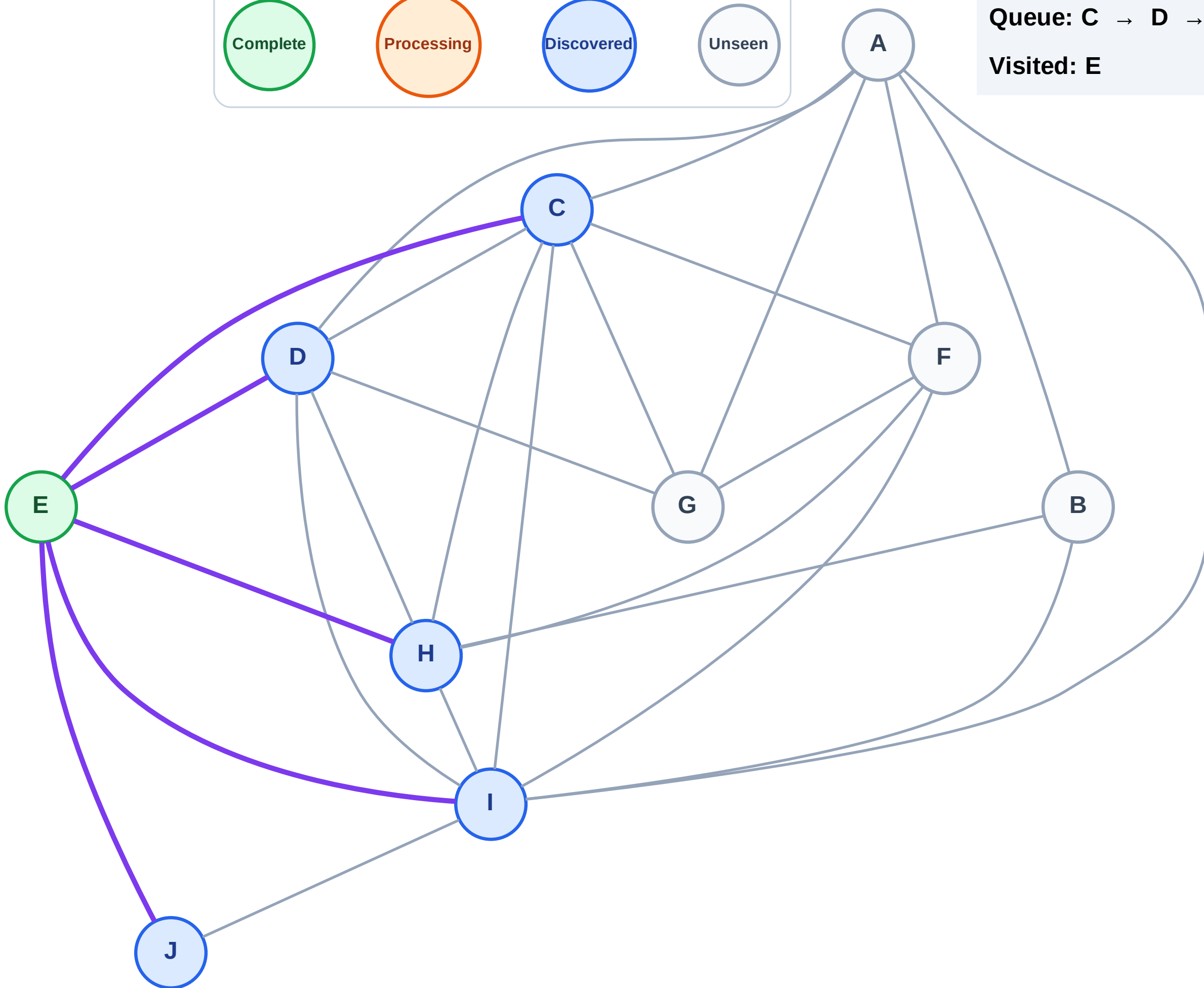
BFS Traversal

Step 3: Discover C, D, H, I, J from E

Legend



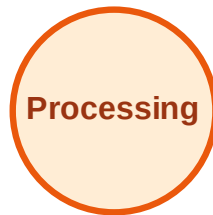
Queue: C → D → H → I → J
Visited: E



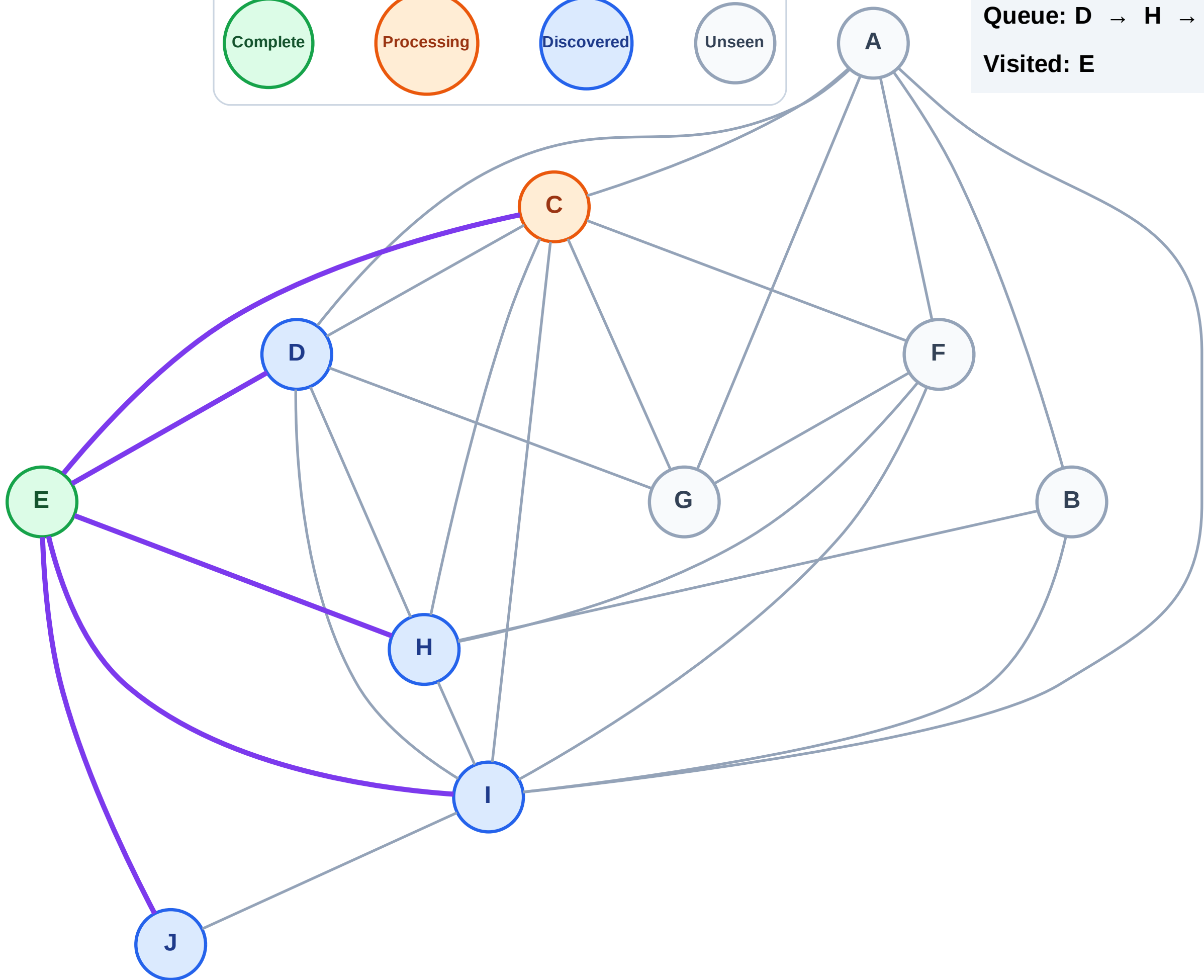
BFS Traversal

Step 4: Process node C

Legend



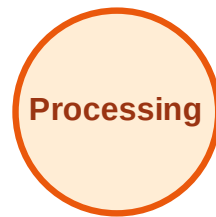
Queue: D → H → I → J
Visited: E



BFS Traversal

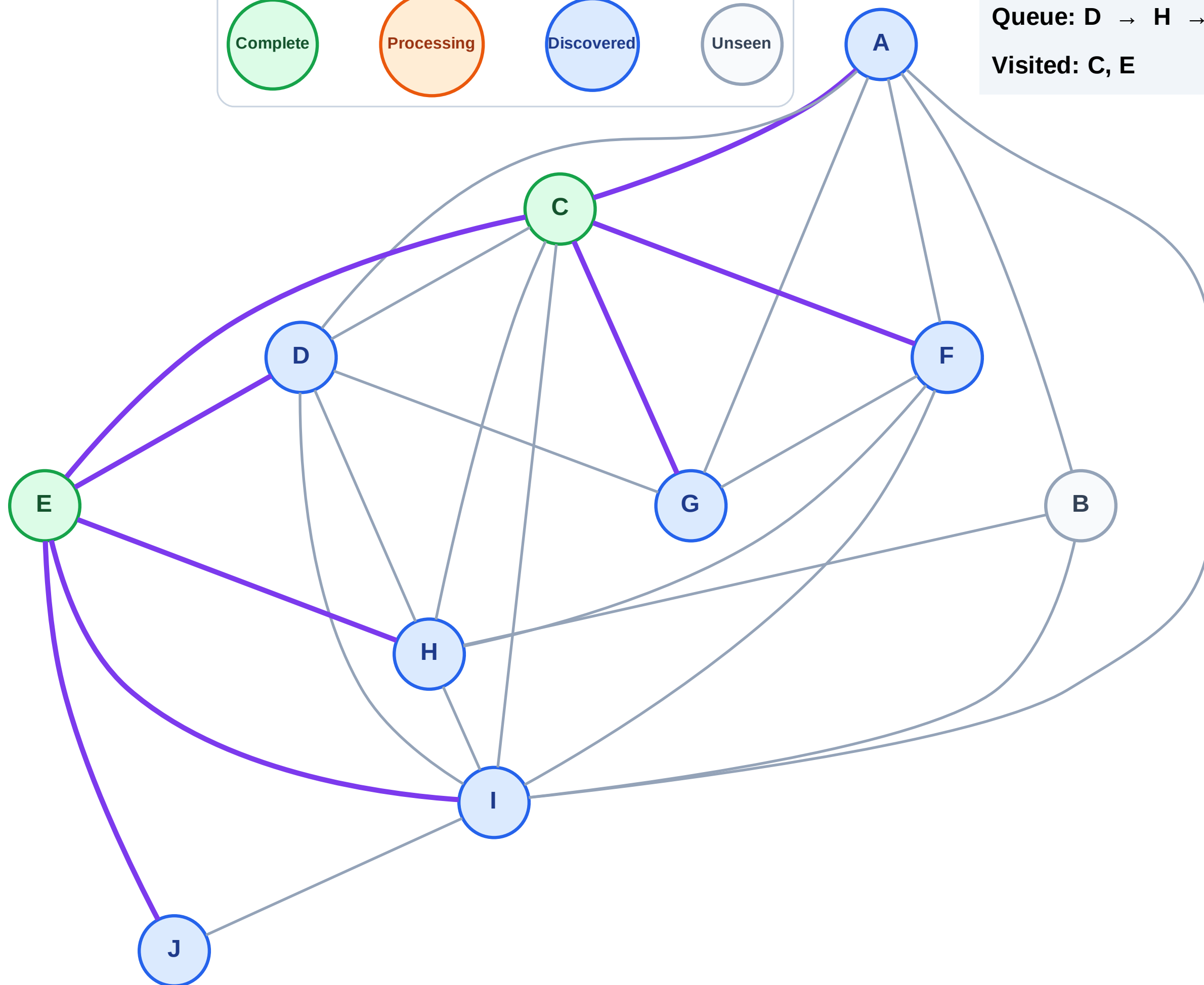
Step 5: Discover A, F, G from C

Legend



Queue: D → H → I → J → A → F → G

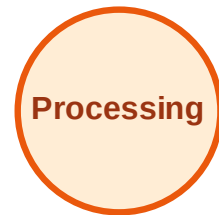
Visited: C, E



BFS Traversal

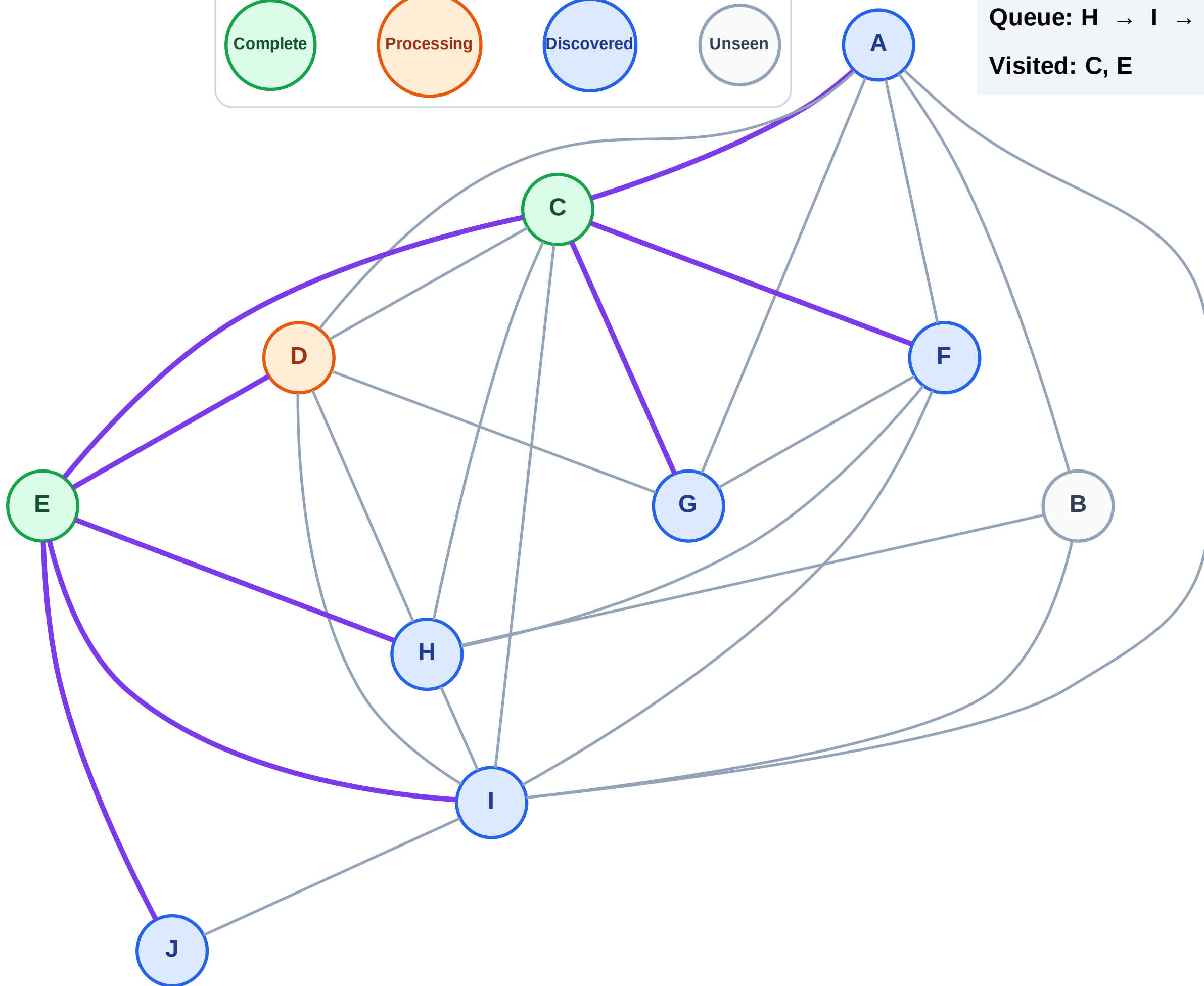
Step 6: Process node D

Legend



Queue: H → I → J → A → F → G

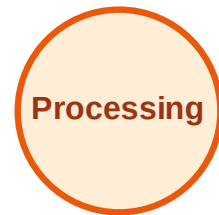
Visited: C, E



BFS Traversal

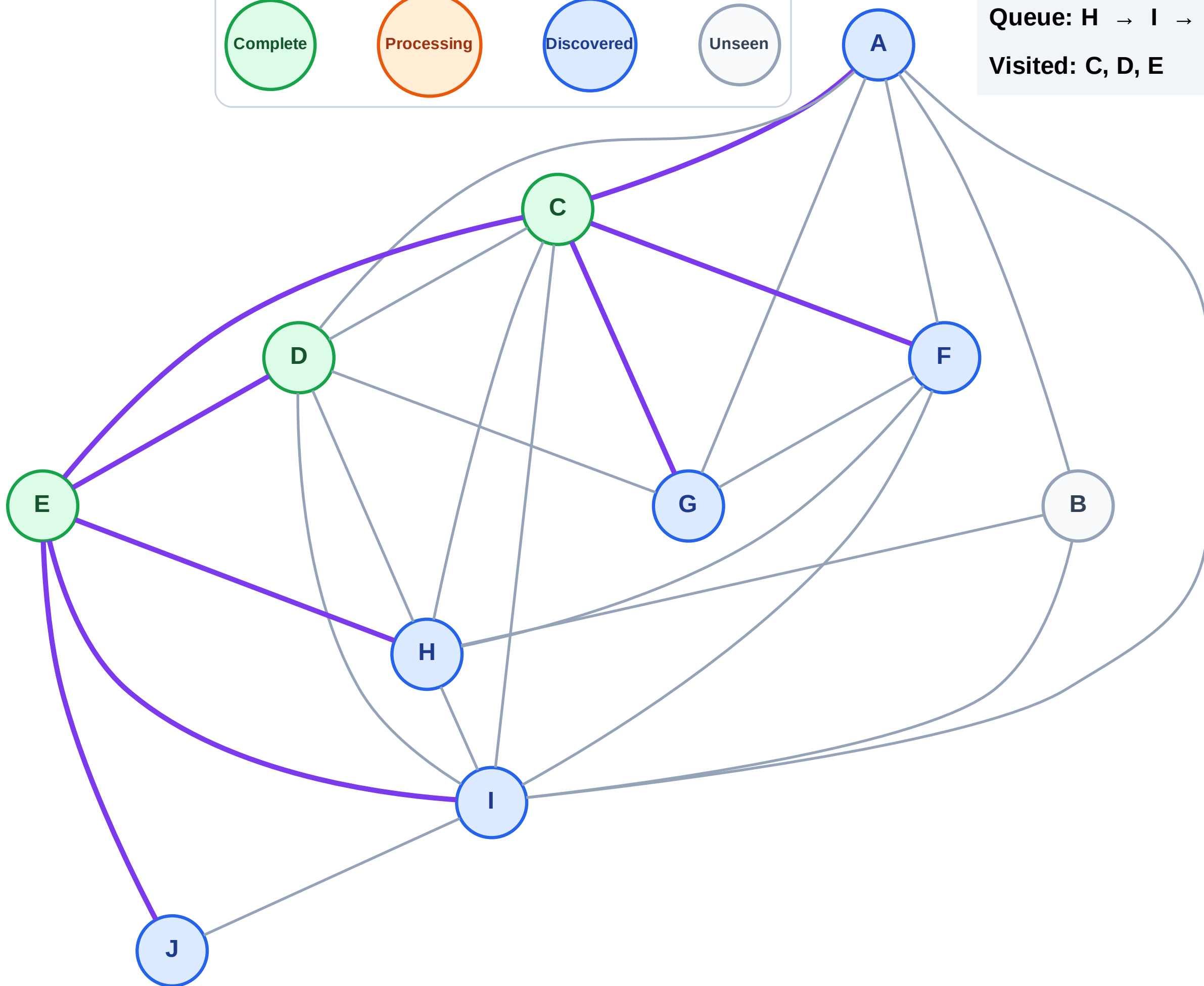
Step 7: No new neighbors from D

Legend



Queue: H → I → J → A → F → G

Visited: C, D, E



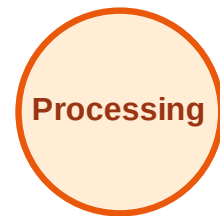
BFS Traversal

Step 8: Process node H

Legend



Complete



Processing

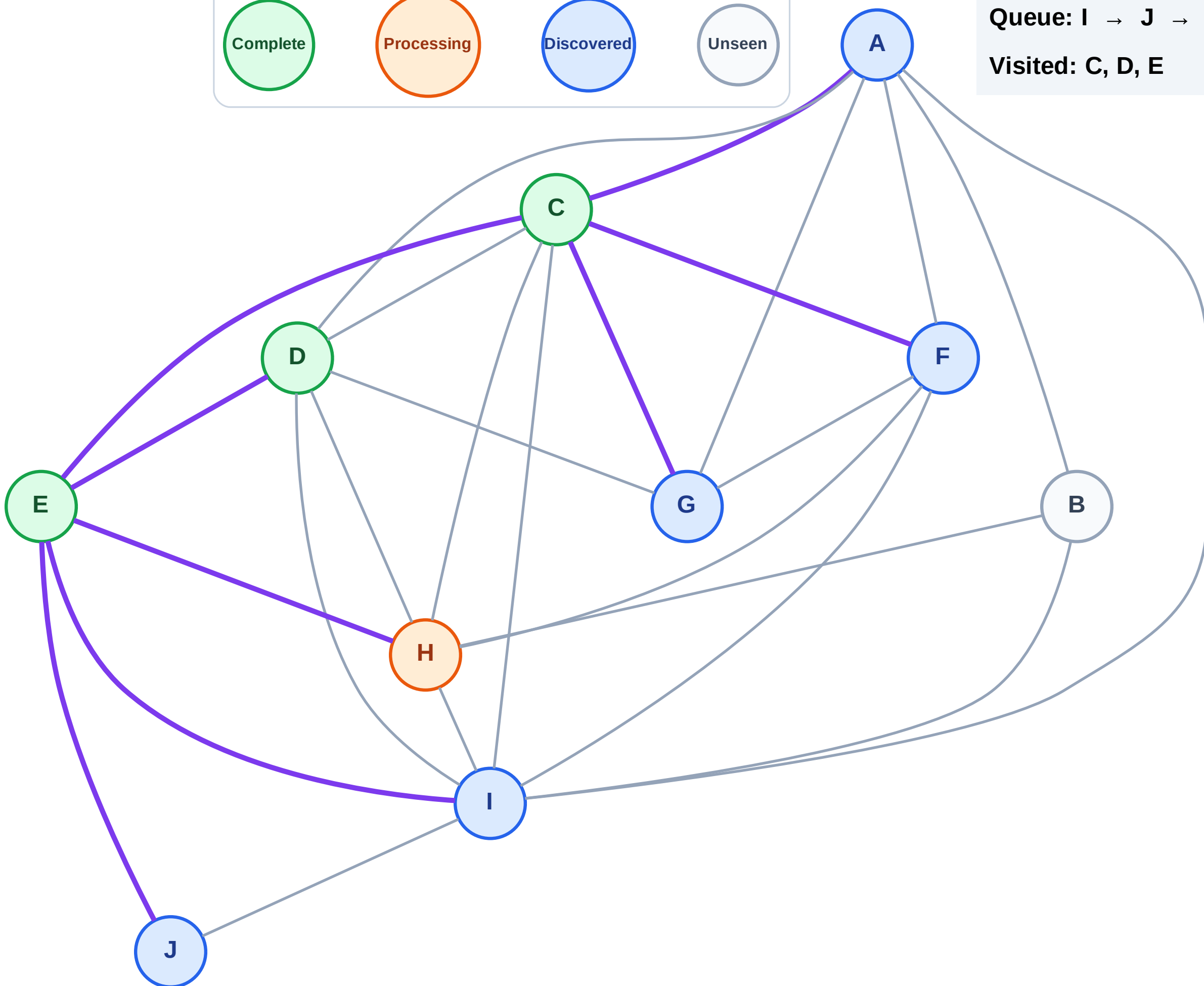


Discovered



Unseen

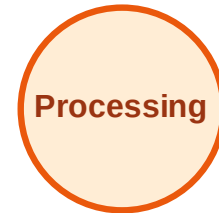
Queue: I → J → A → F → G
Visited: C, D, E



BFS Traversal

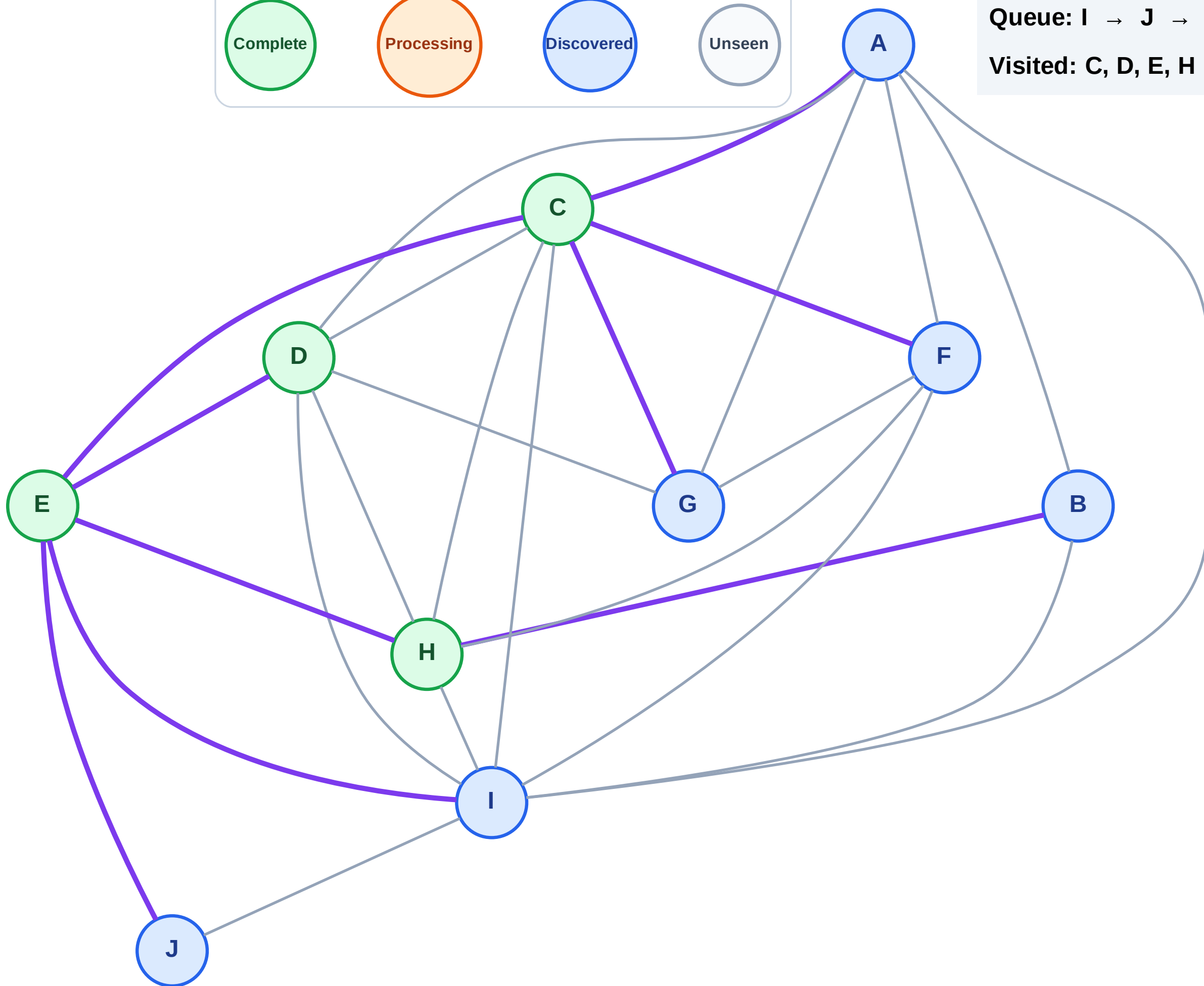
Step 9: Discover B from H

Legend



Queue: I → J → A → F → G → B

Visited: C, D, E, H



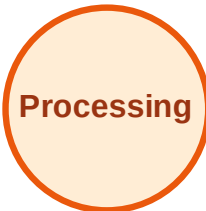
BFS Traversal

Step 10: Process node I

Legend



Processing



Discovered

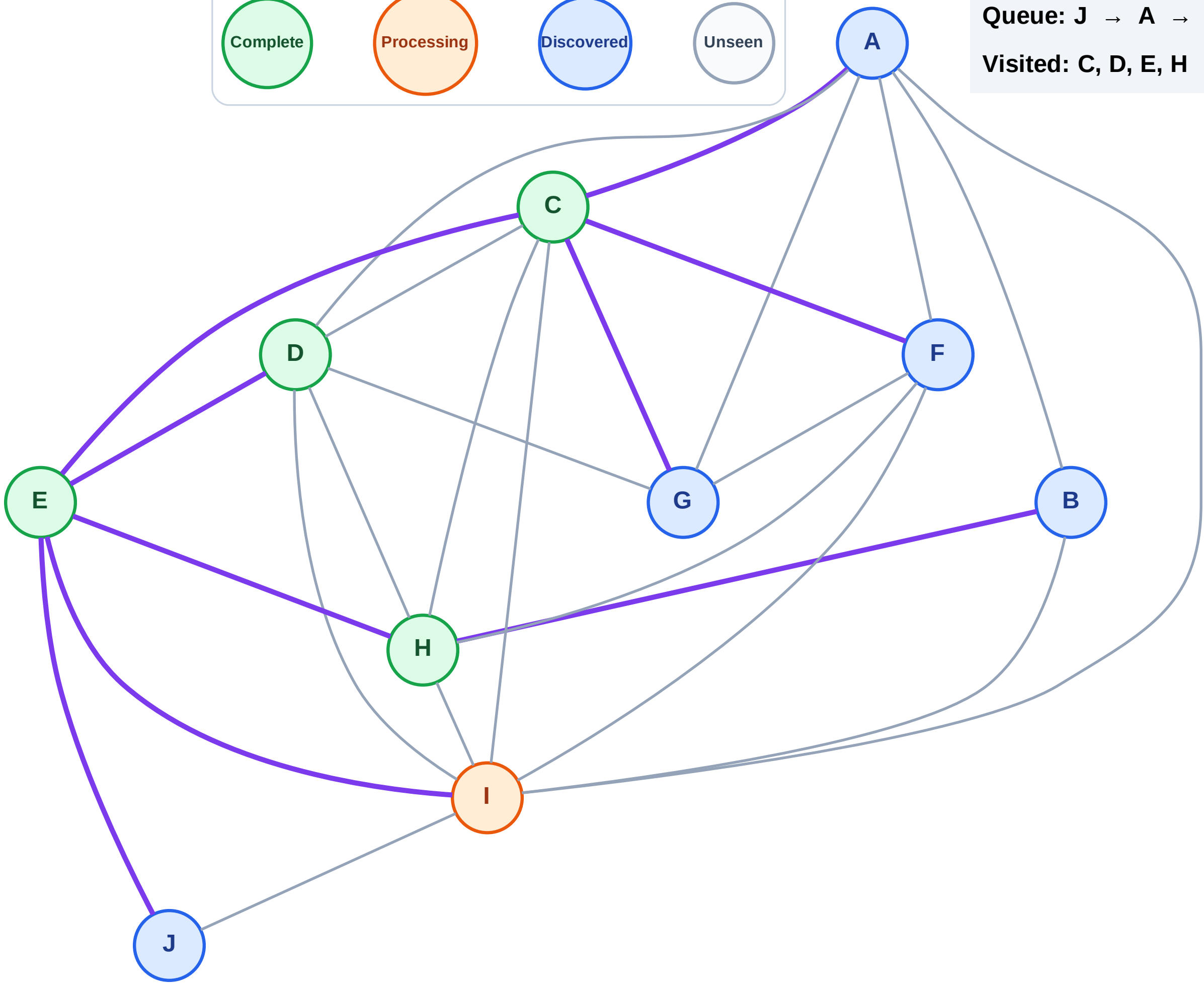


Unseen



Queue: J → A → F → G → B

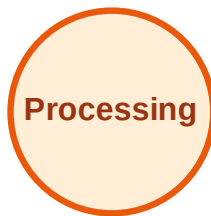
Visited: C, D, E, H



BFS Traversal

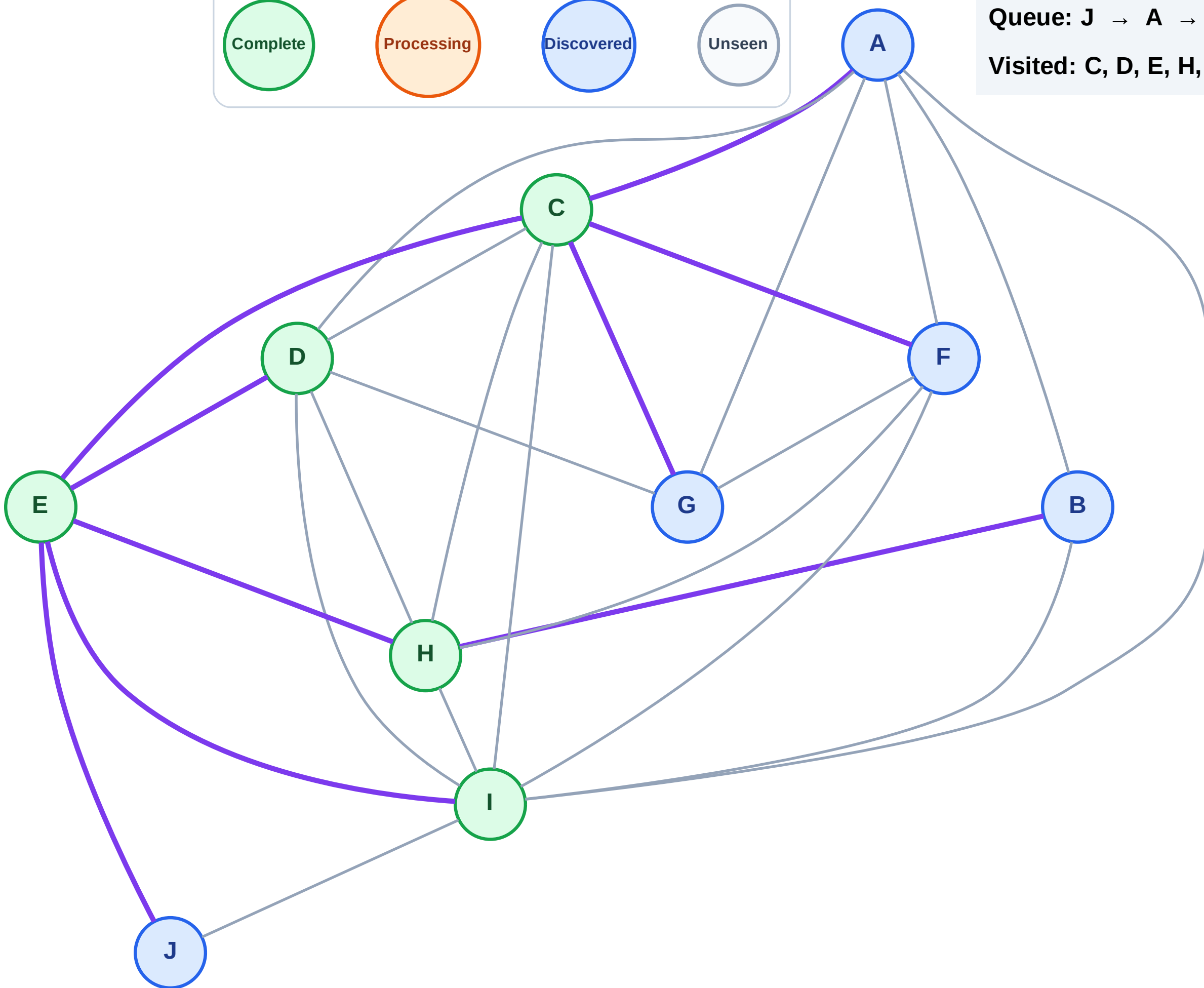
Step 11: No new neighbors from I

Legend



Queue: J → A → F → G → B

Visited: C, D, E, H, I



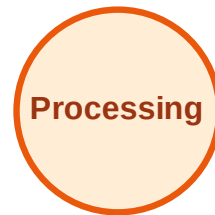
BFS Traversal

Step 12: Process node J

Legend



Complete



Processing

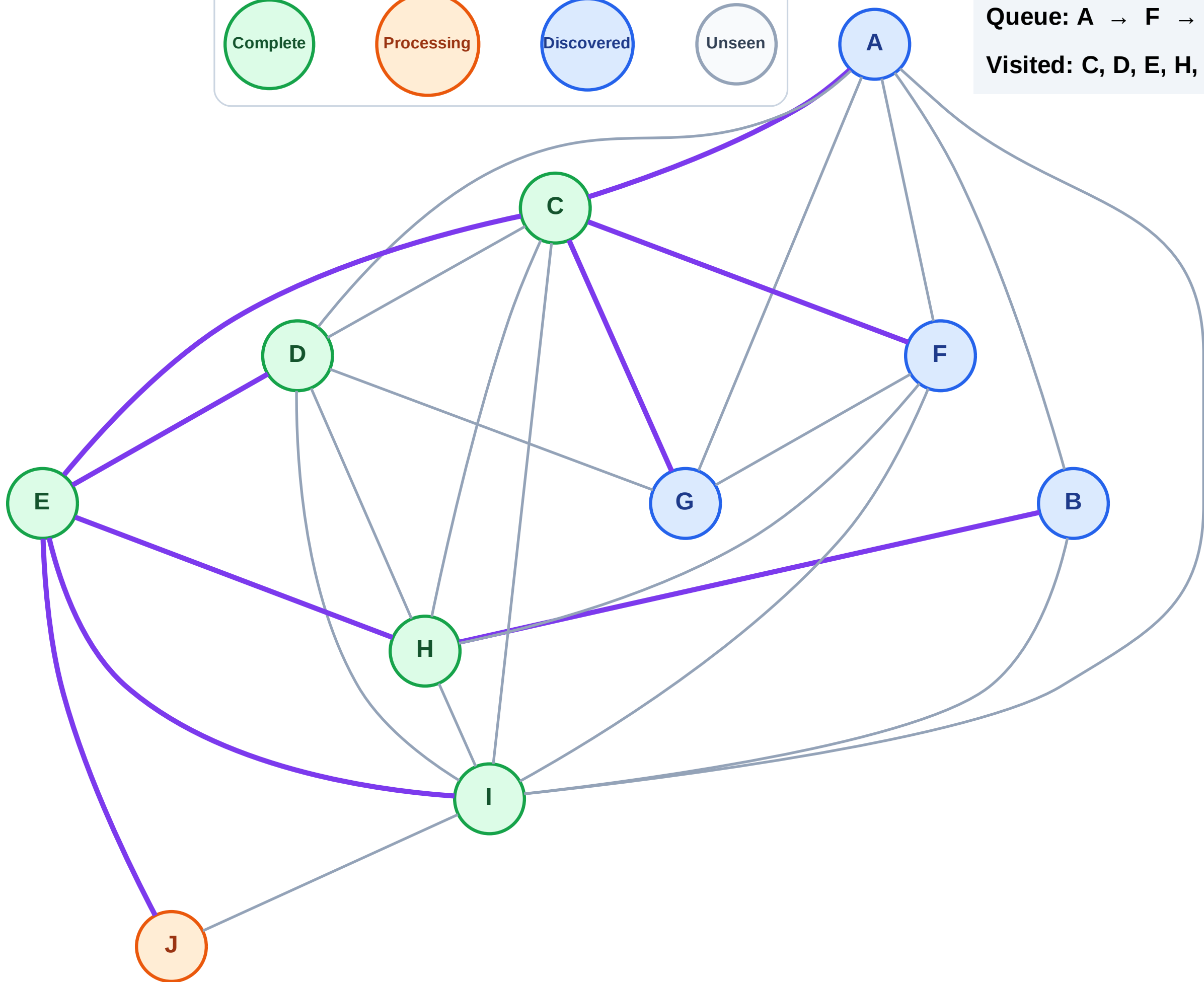


Discovered



Unseen

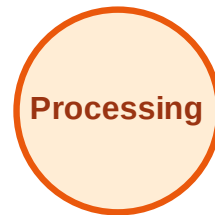
Queue: A → F → G → B
Visited: C, D, E, H, I



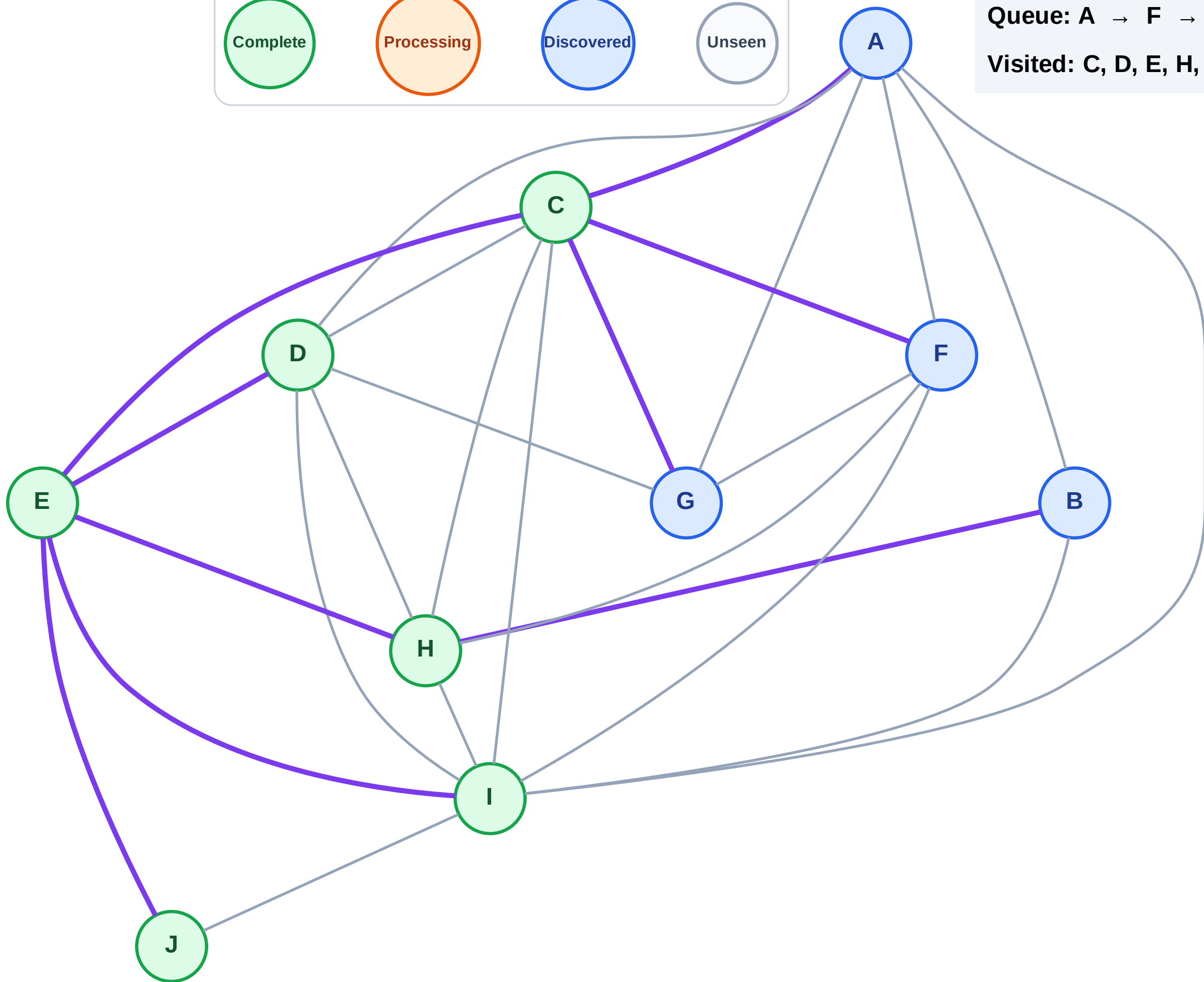
BFS Traversal

Step 13: No new neighbors from J

Legend



Queue: A → F → G → B
Visited: C, D, E, H, I, J



BFS Traversal

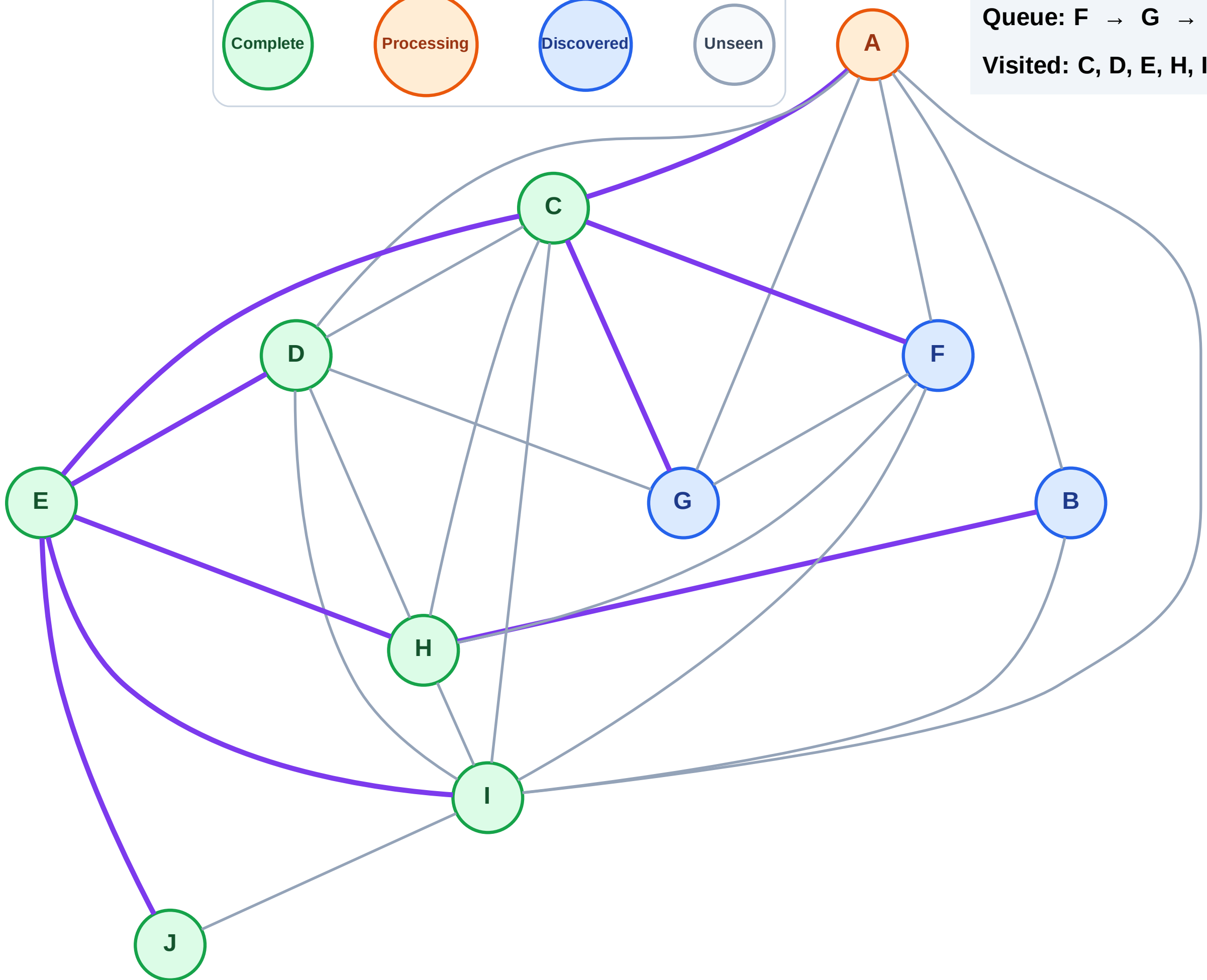
Step 14: Process node A

Legend



Queue: F → G → B

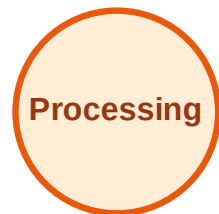
Visited: C, D, E, H, I, J



BFS Traversal

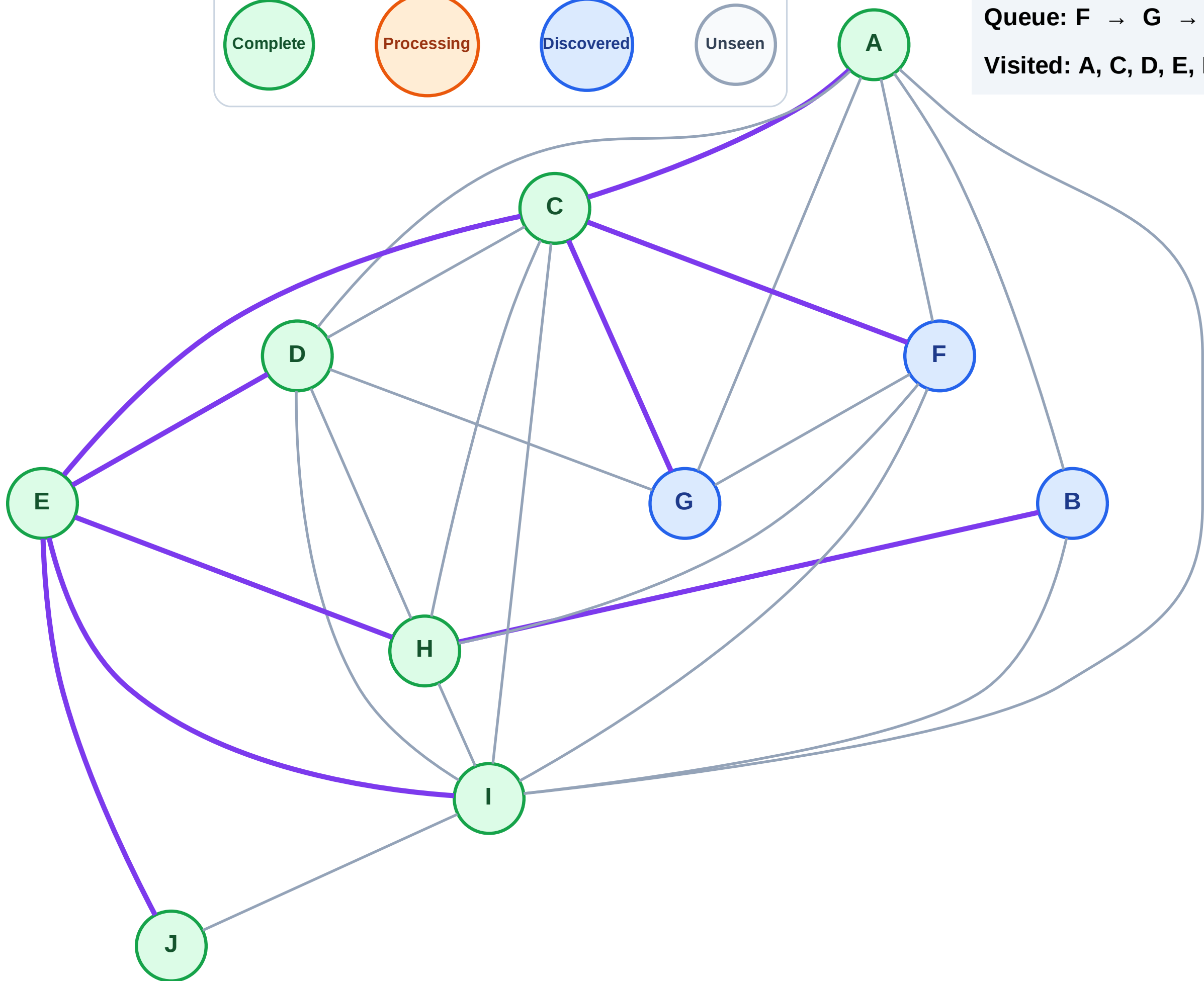
Step 15: No new neighbors from A

Legend



Queue: F → G → B

Visited: A, C, D, E, H, I, J



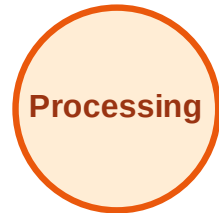
BFS Traversal

Step 16: Process node F

Legend



Complete



Processing



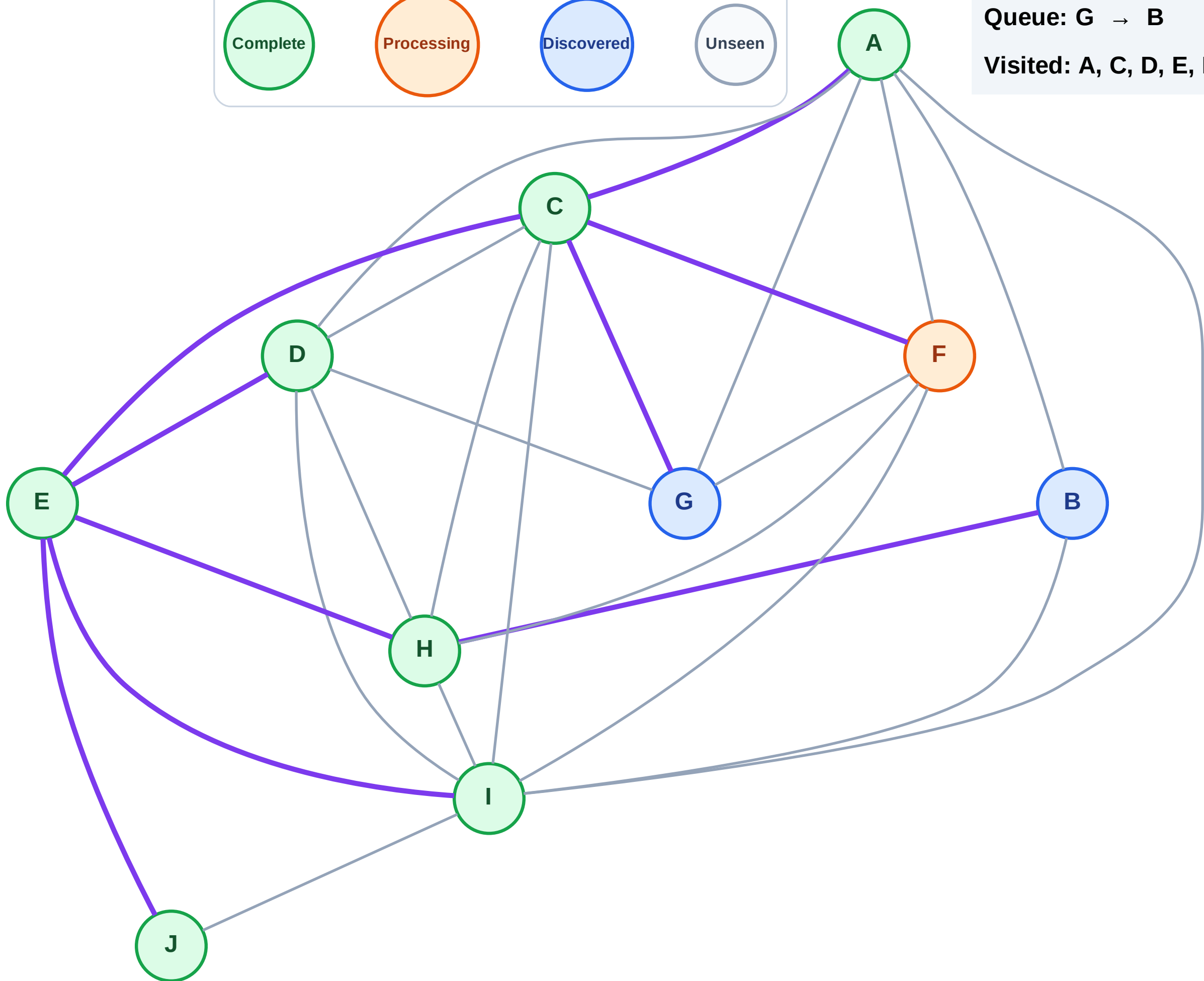
Discovered



Unseen

Queue: G → B

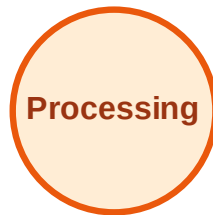
Visited: A, C, D, E, H, I, J



BFS Traversal

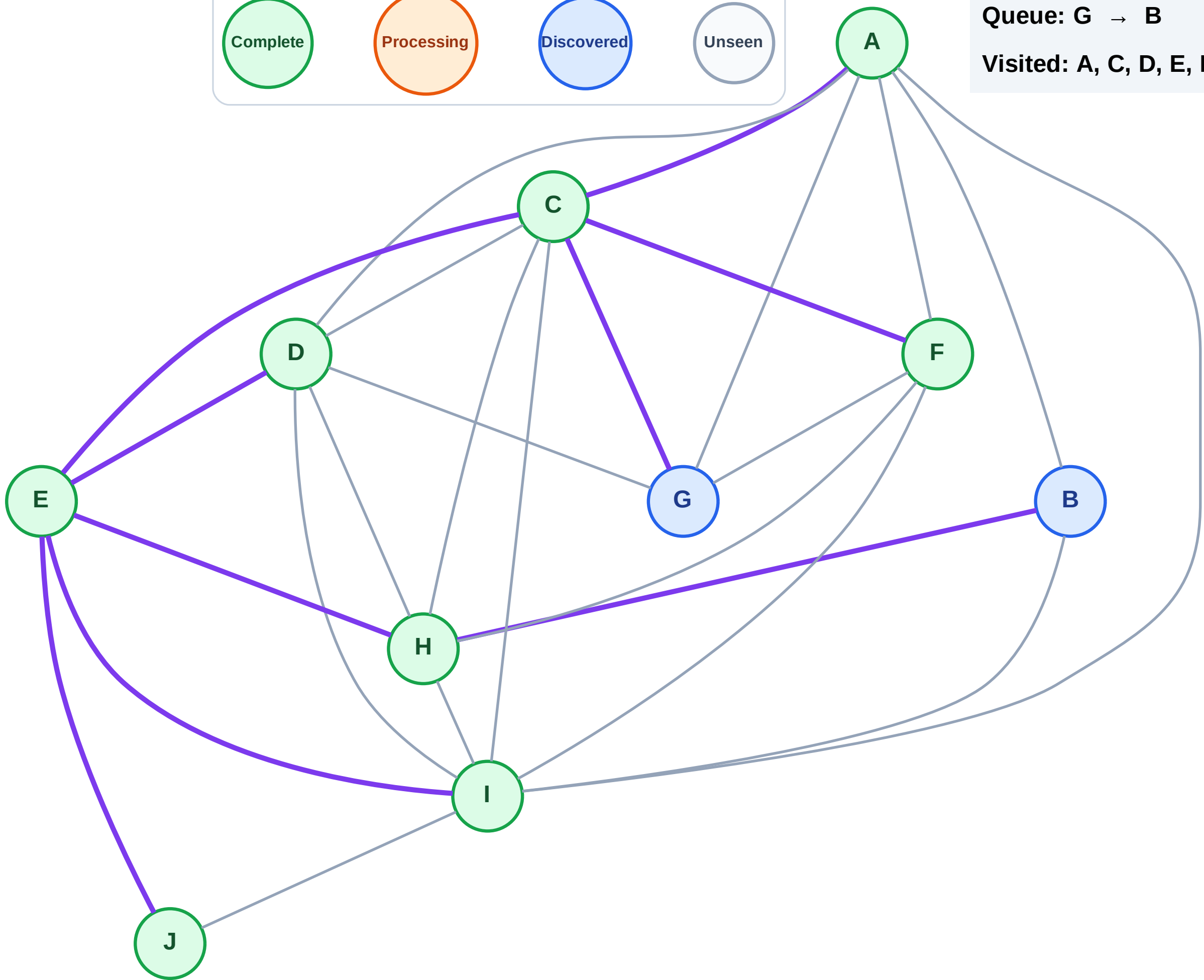
Step 17: No new neighbors from F

Legend



Queue: G → B

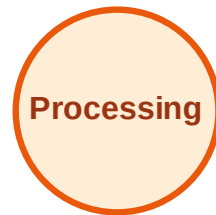
Visited: A, C, D, E, F, H, I, J



BFS Traversal

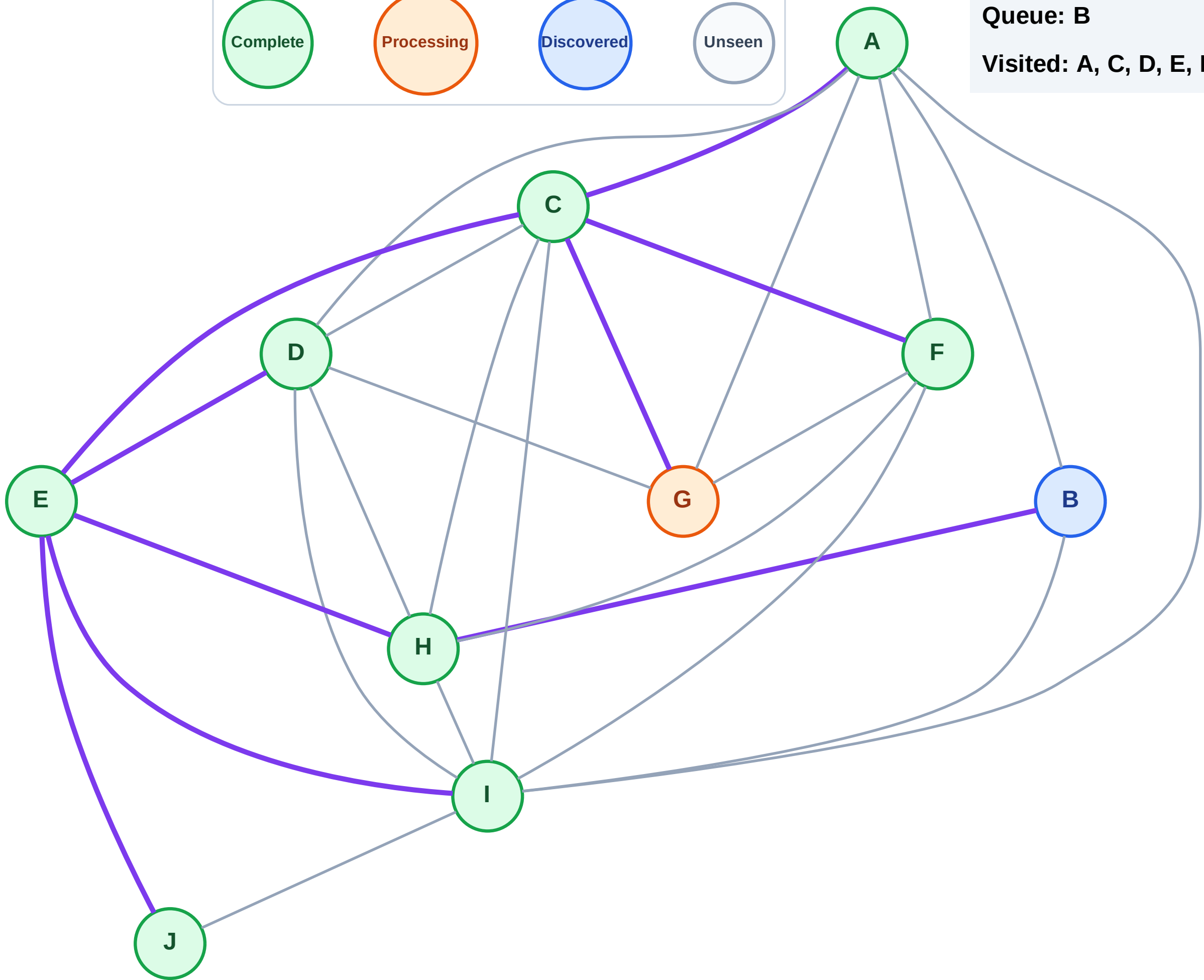
Step 18: Process node G

Legend



Queue: B

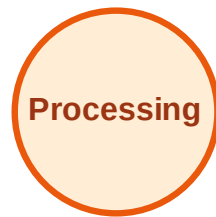
Visited: A, C, D, E, F, H, I, J



BFS Traversal

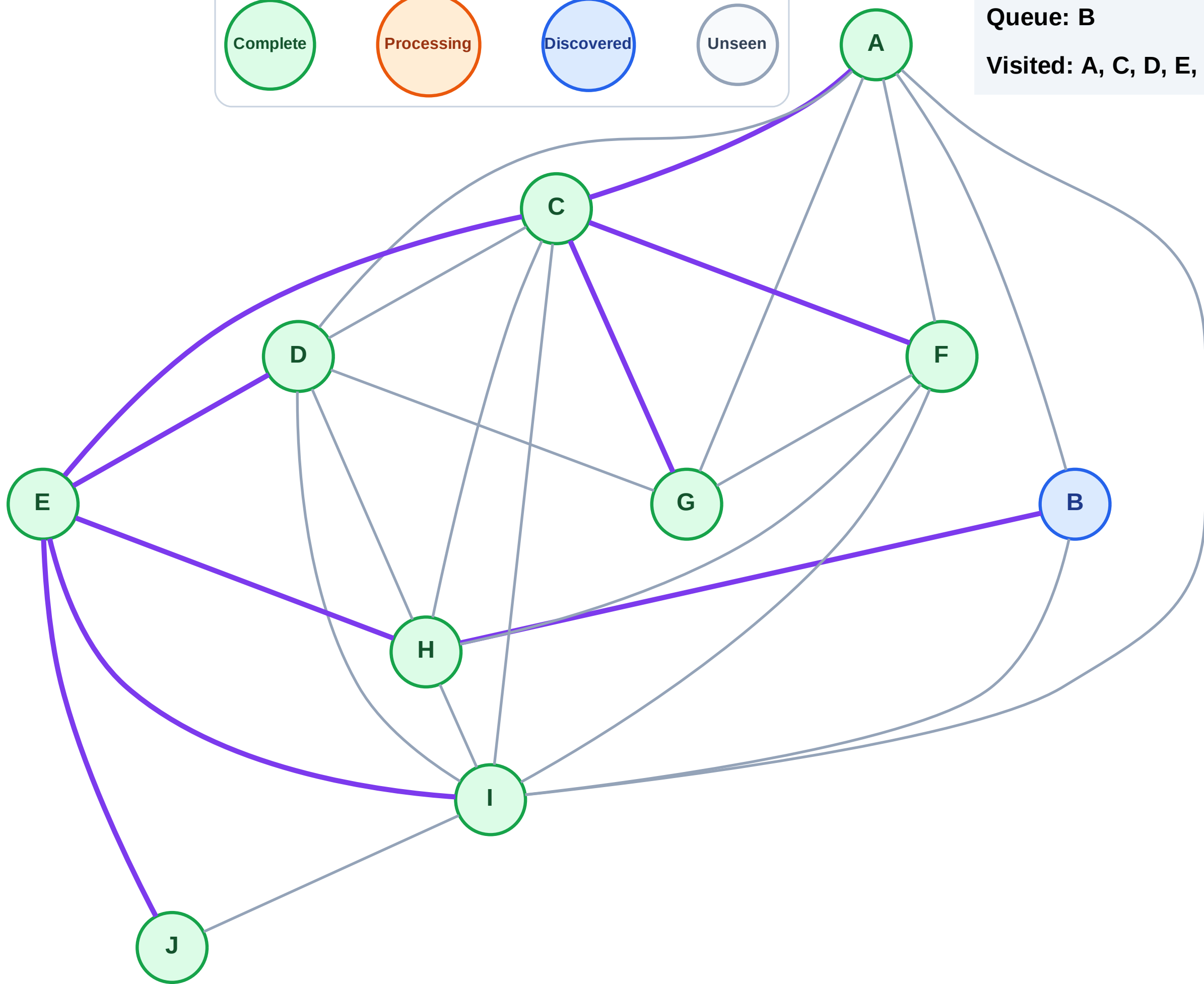
Step 19: No new neighbors from G

Legend



Queue: B

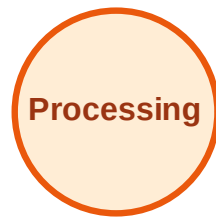
Visited: A, C, D, E, F, G, H, I, J



BFS Traversal

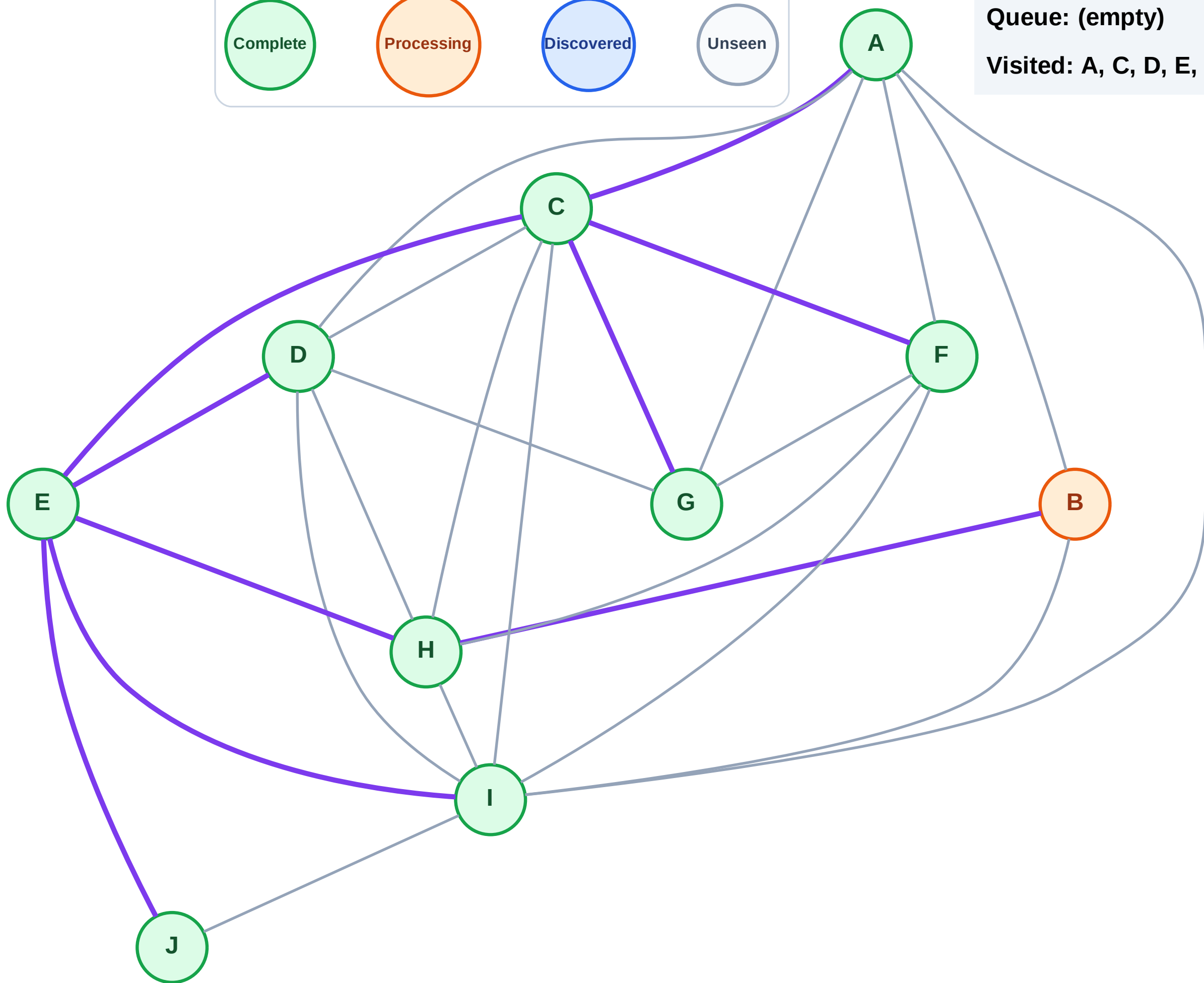
Step 20: Process node B

Legend



Queue: (empty)

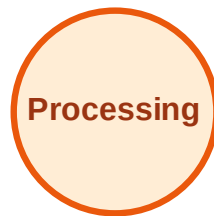
Visited: A, C, D, E, F, G, H, I, J



BFS Traversal

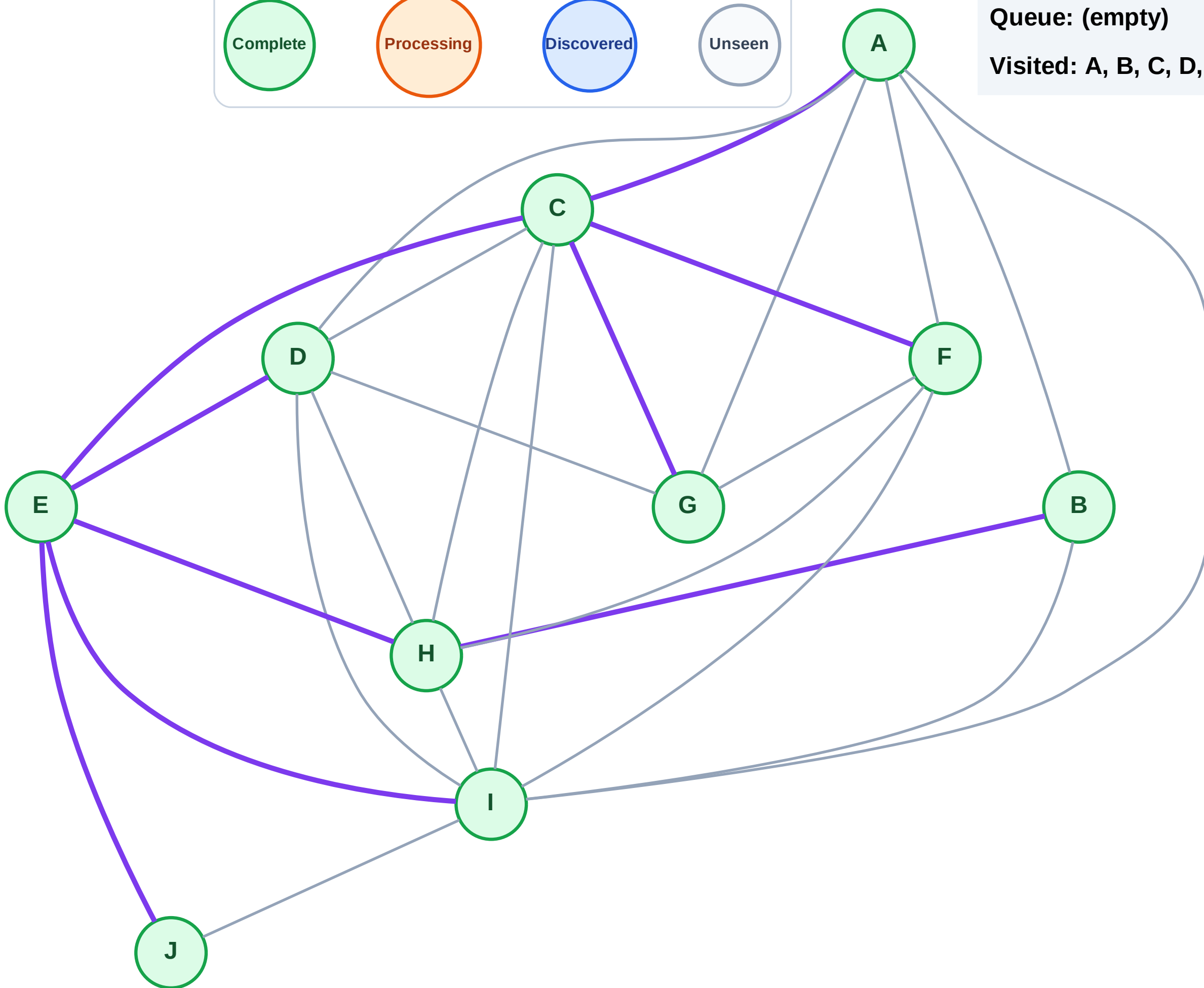
Step 21: No new neighbors from B

Legend



Queue: (empty)

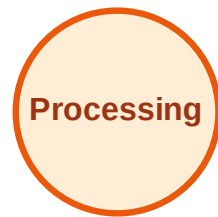
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

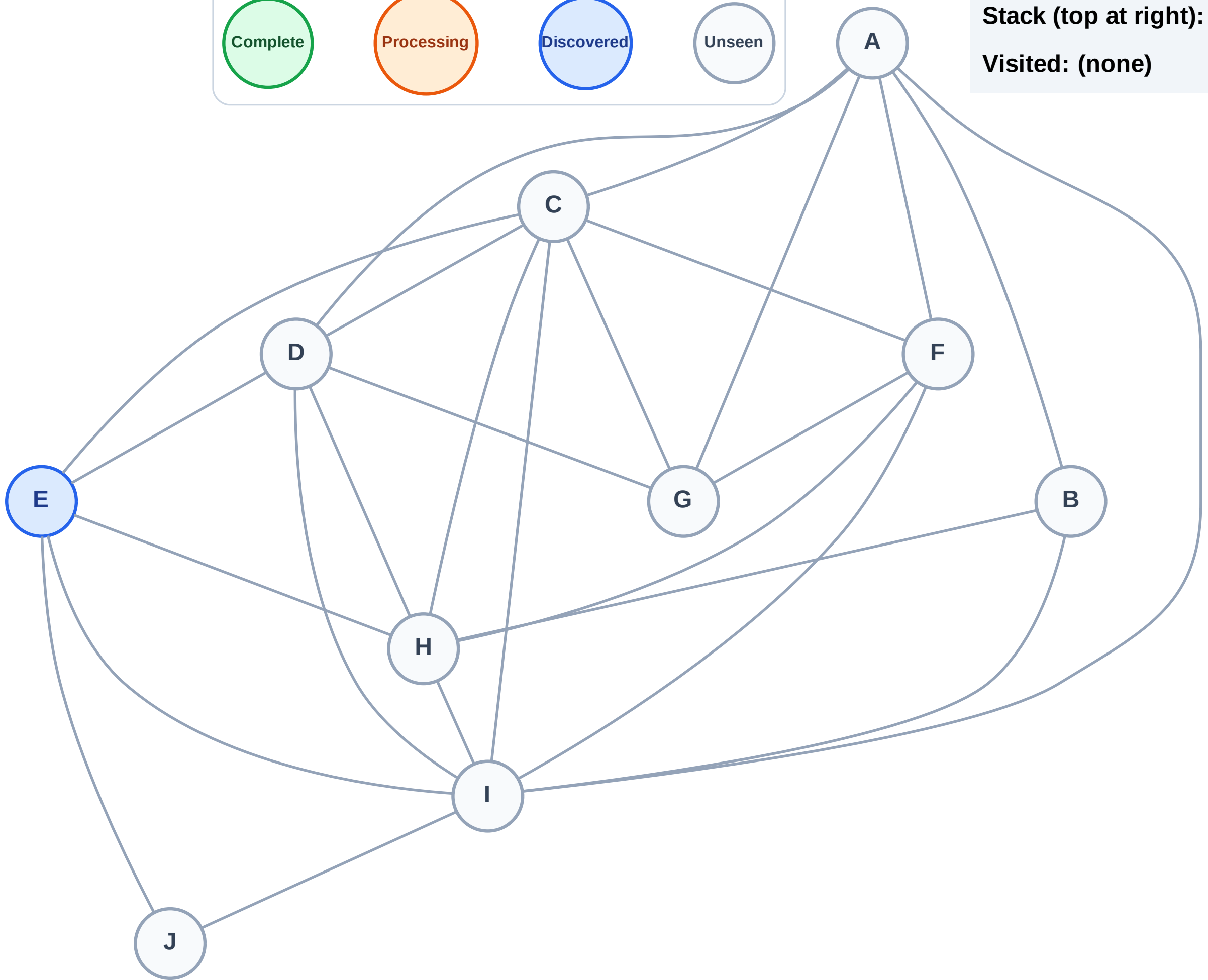
Step 1: Start at E

Legend



Stack (top at right): E

Visited: (none)



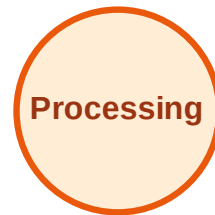
DFS Traversal

Step 2: Process node E

Legend



Complete



Processing



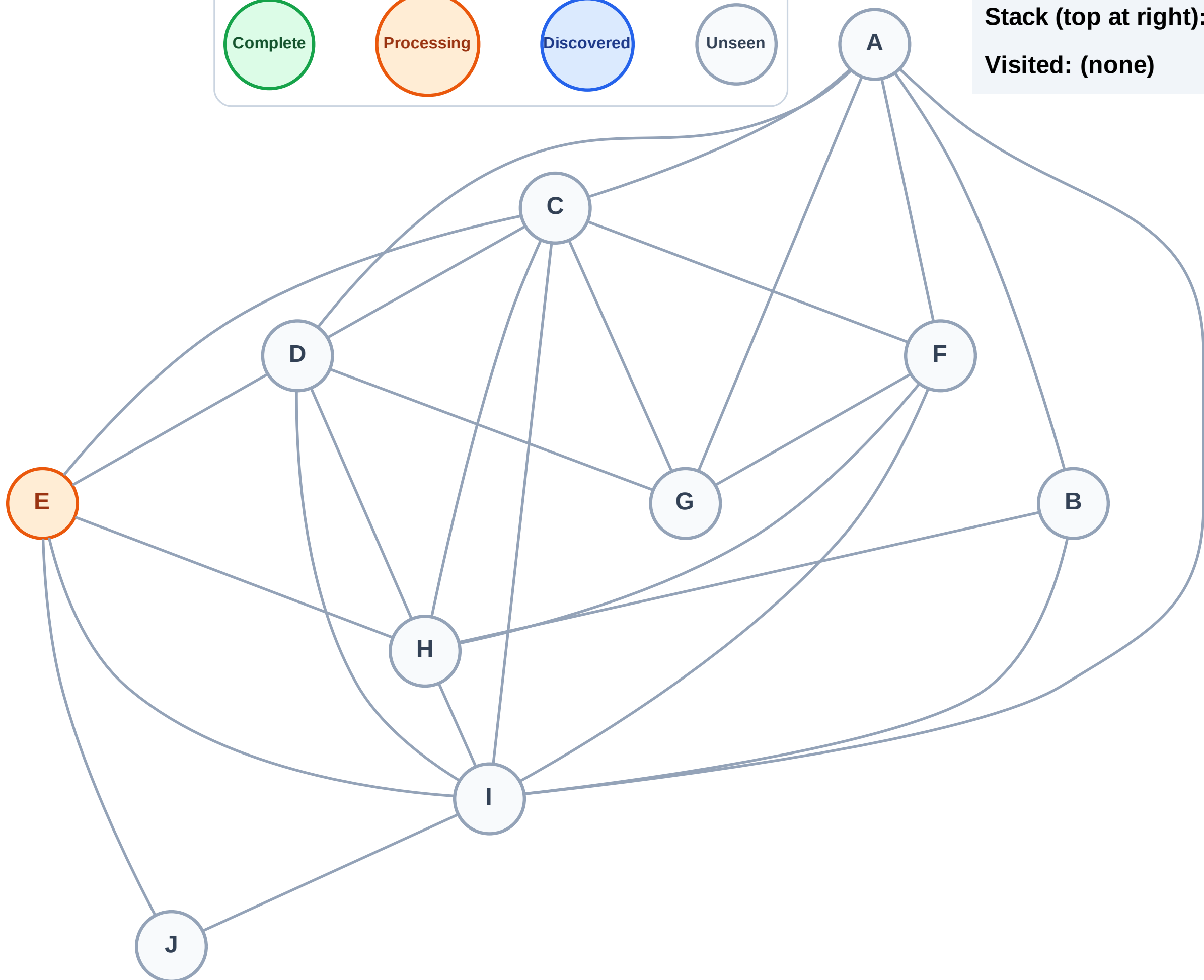
Discovered



Unseen

Stack (top at right): (empty)

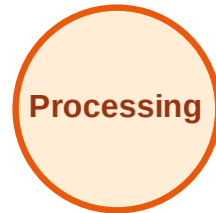
Visited: (none)



DFS Traversal

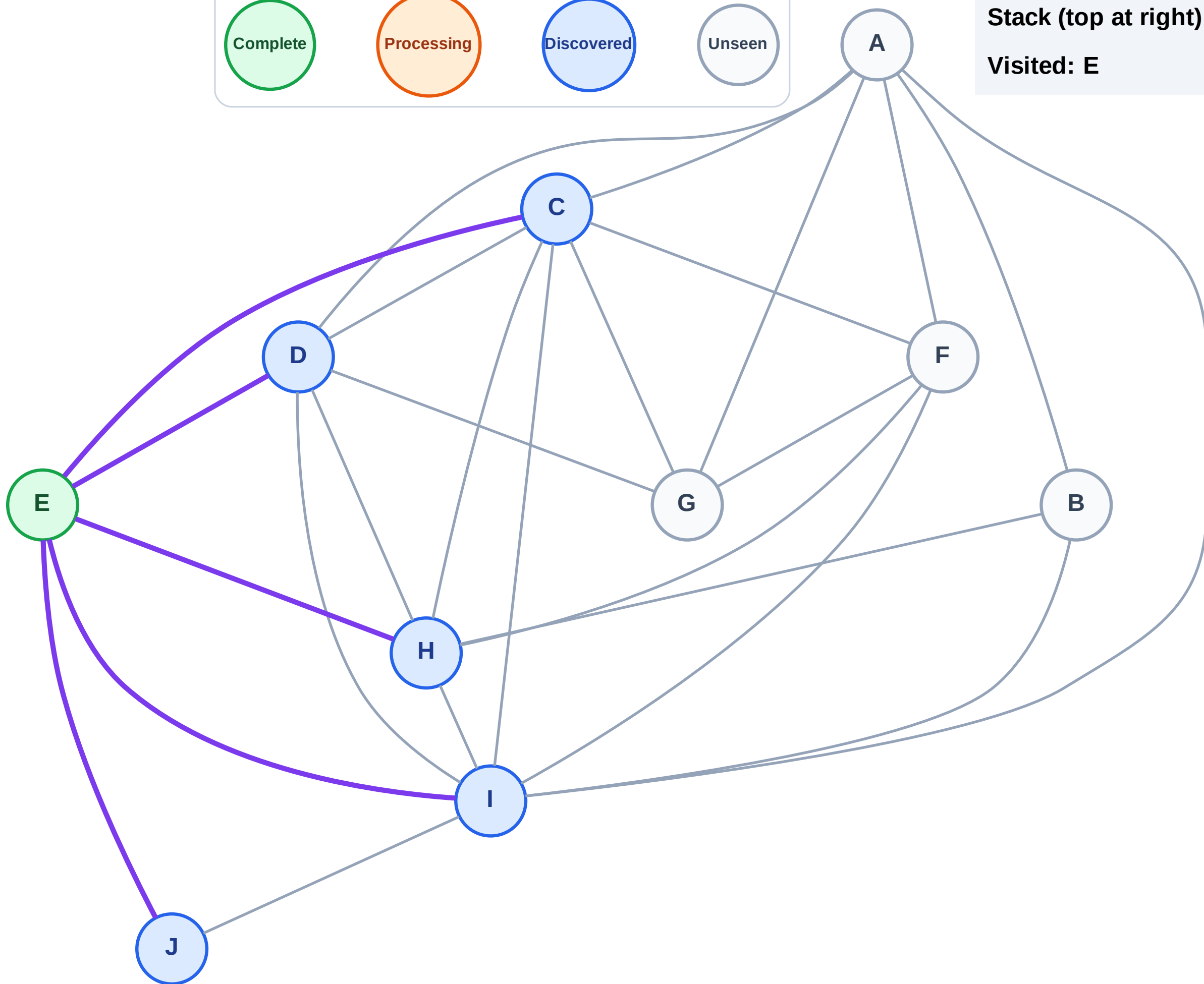
Step 3: Discover J, I, H, D, C from E

Legend



Stack (top at right): J | I | H | D | C

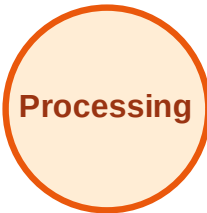
Visited: E



DFS Traversal

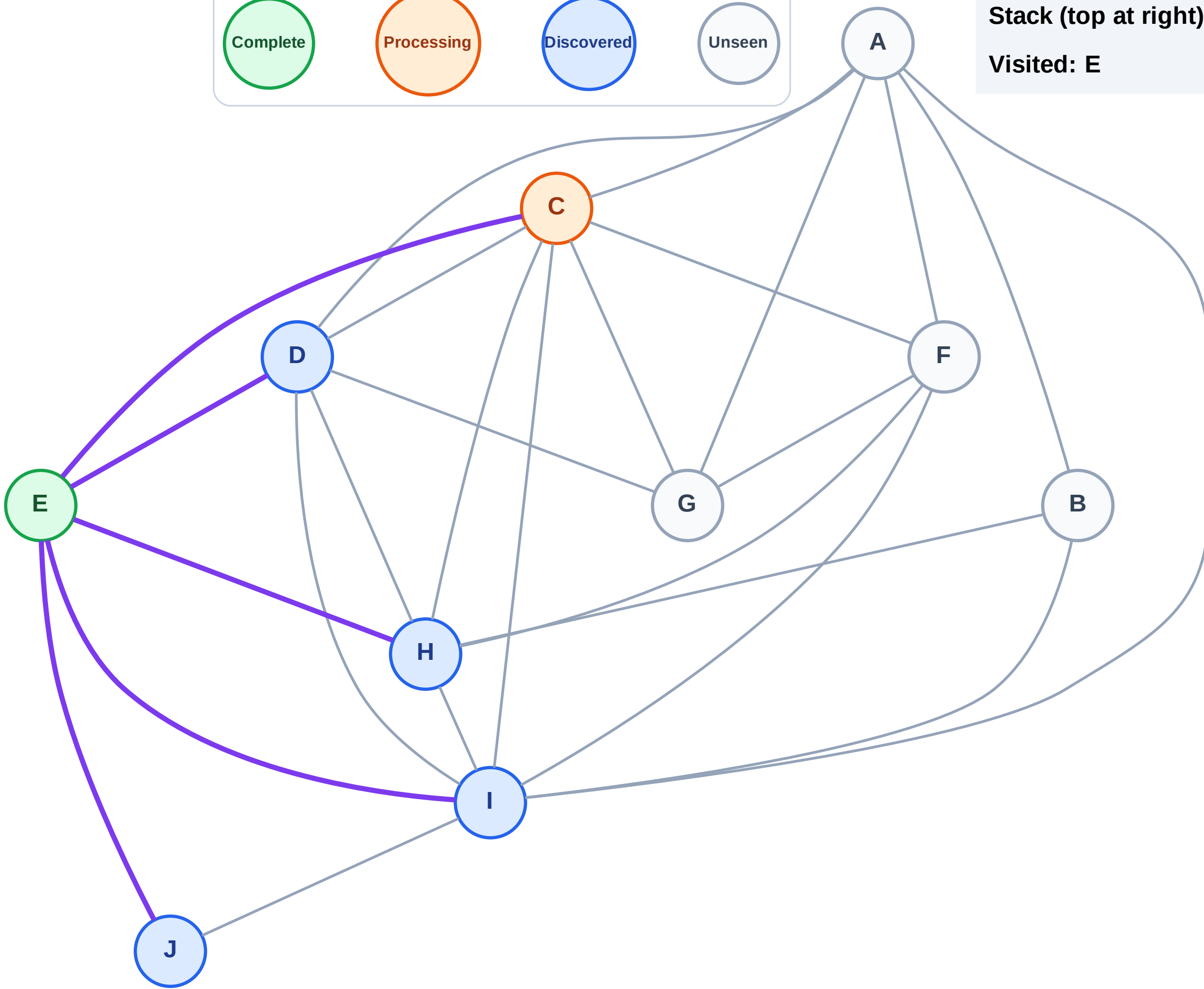
Step 4: Process node C

Legend



Stack (top at right): J | I | H | D

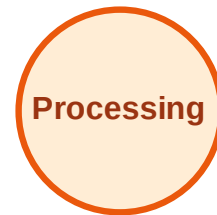
Visited: E



DFS Traversal

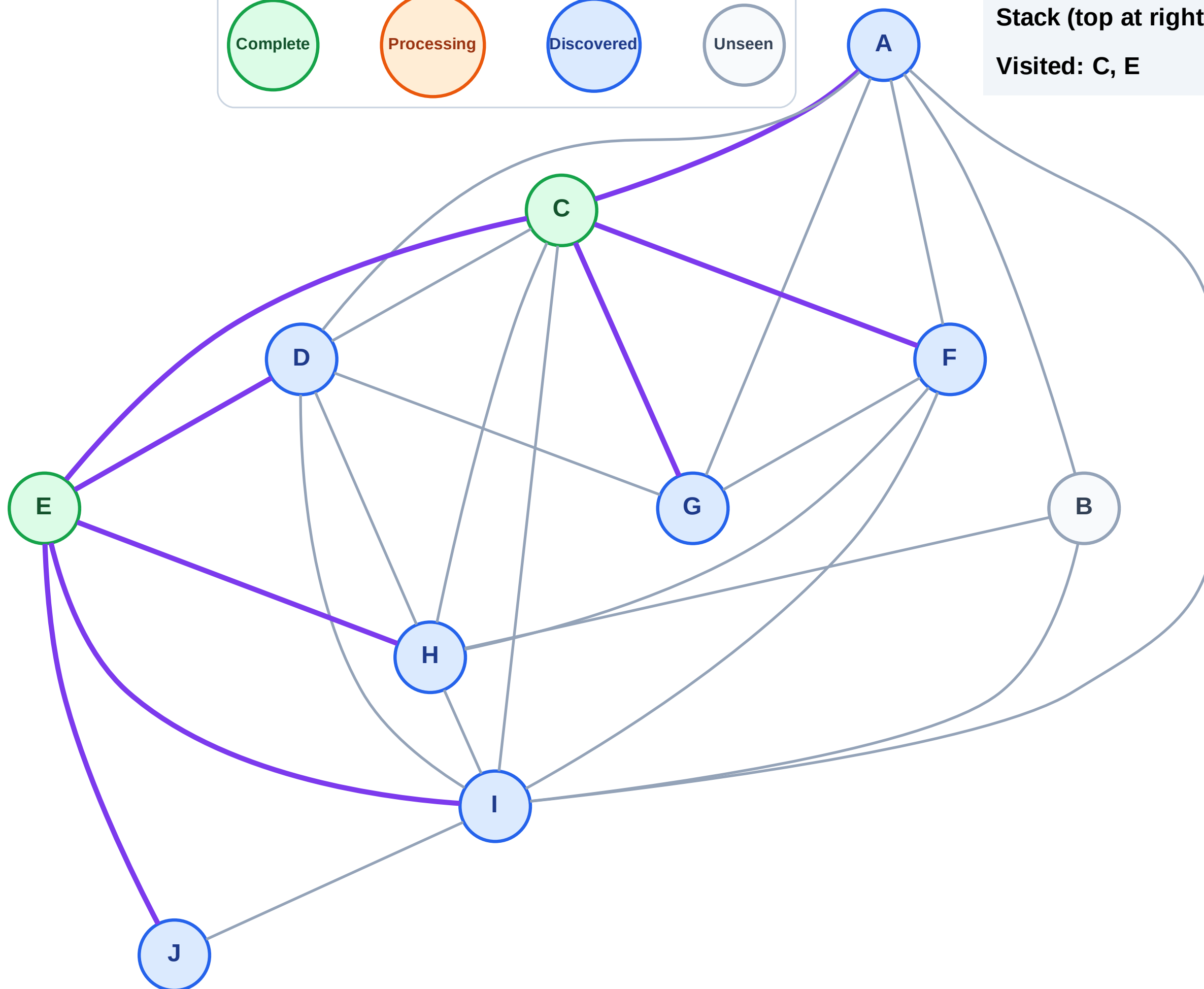
Step 5: Discover G, F, A from C

Legend



Stack (top at right): J | I | H | D | G | F | A

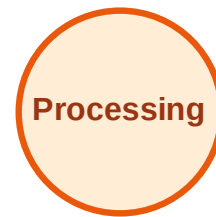
Visited: C, E



DFS Traversal

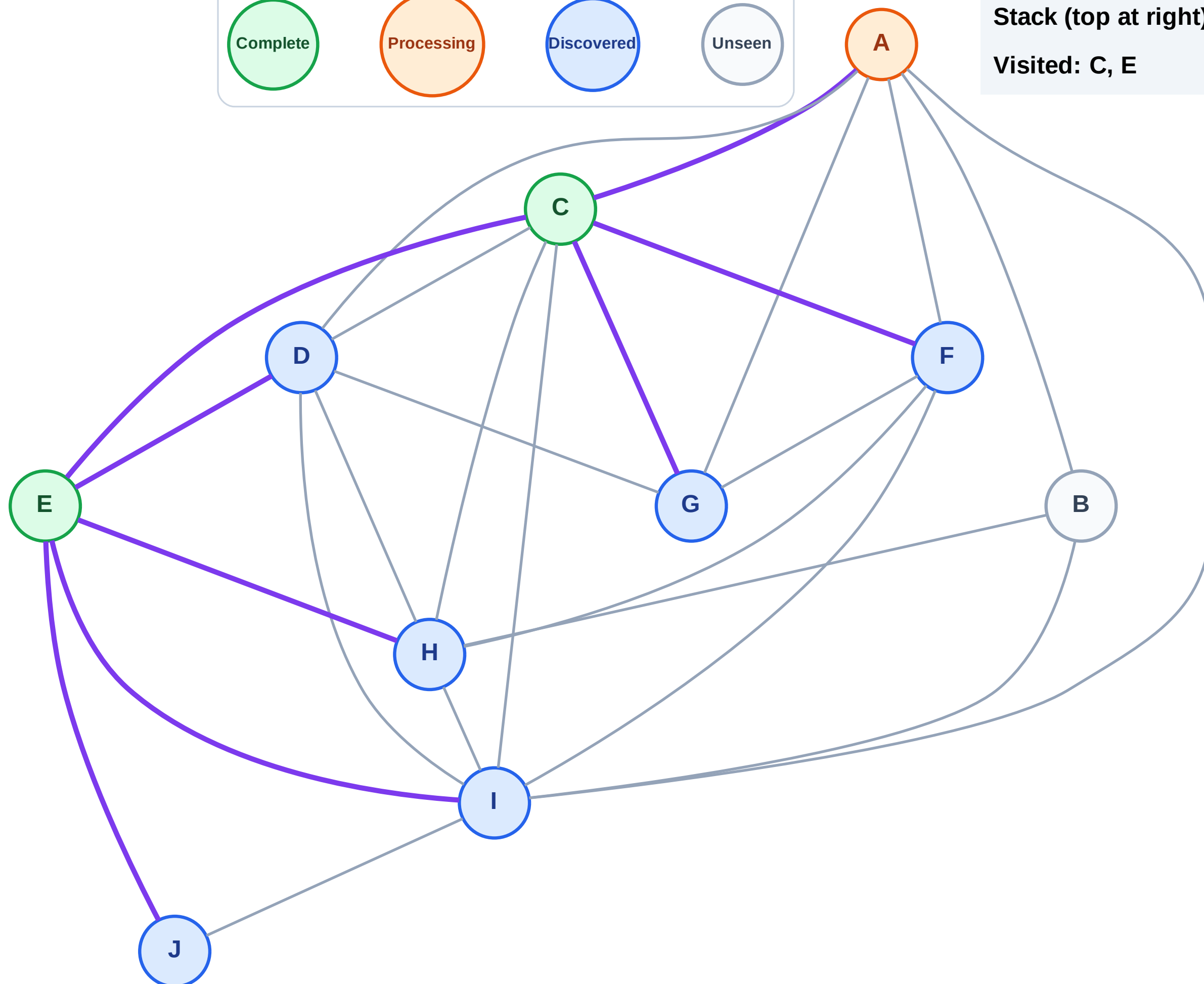
Step 6: Process node A

Legend



Stack (top at right): J | I | H | D | G | F

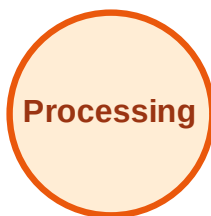
Visited: C, E



DFS Traversal

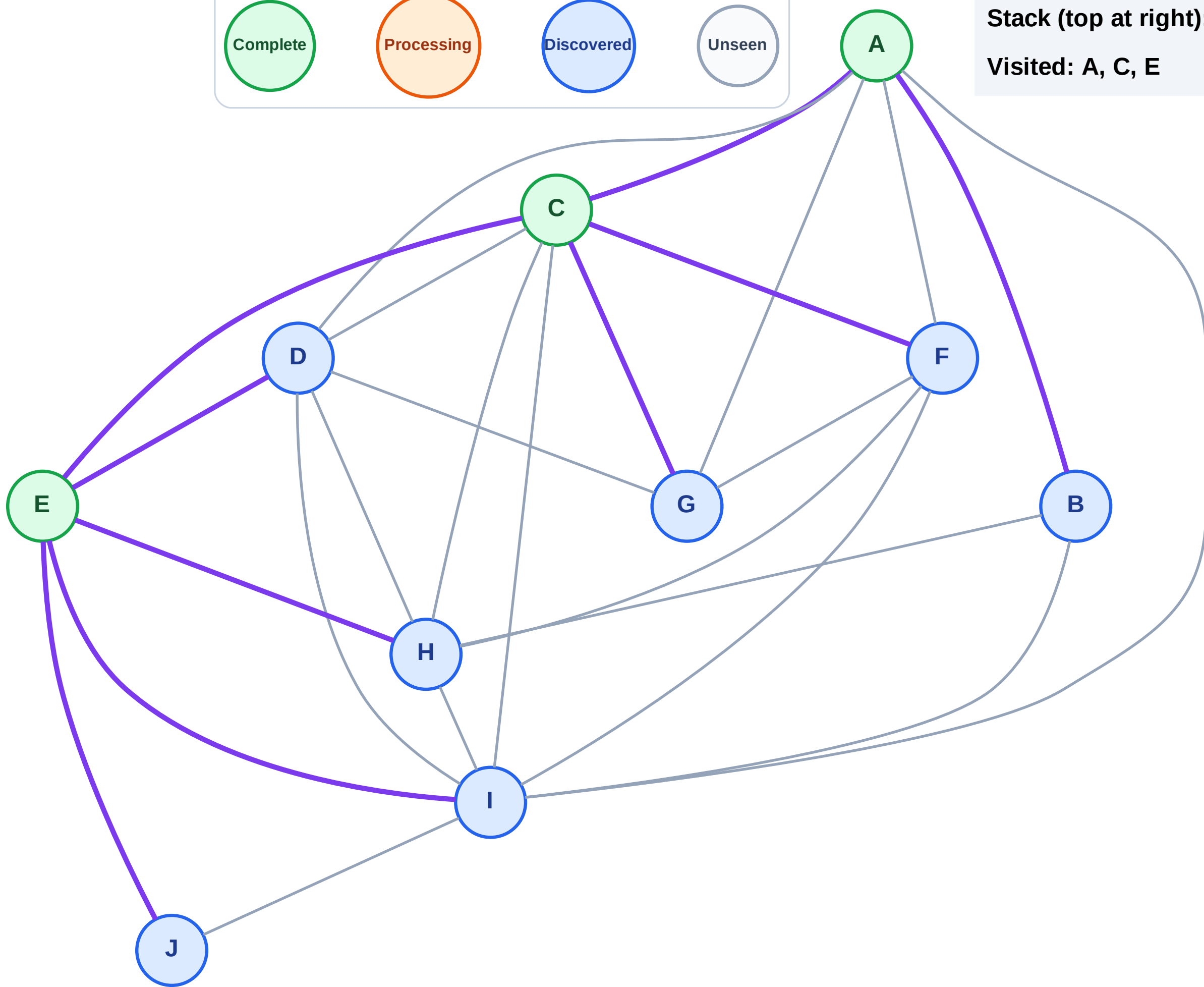
Step 7: Discover B from A

Legend



Stack (top at right): J | I | H | D | G | F | B

Visited: A, C, E



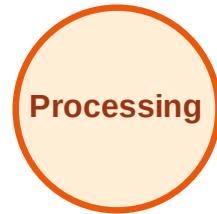
DFS Traversal

Step 8: Process node B

Legend



Complete



Processing



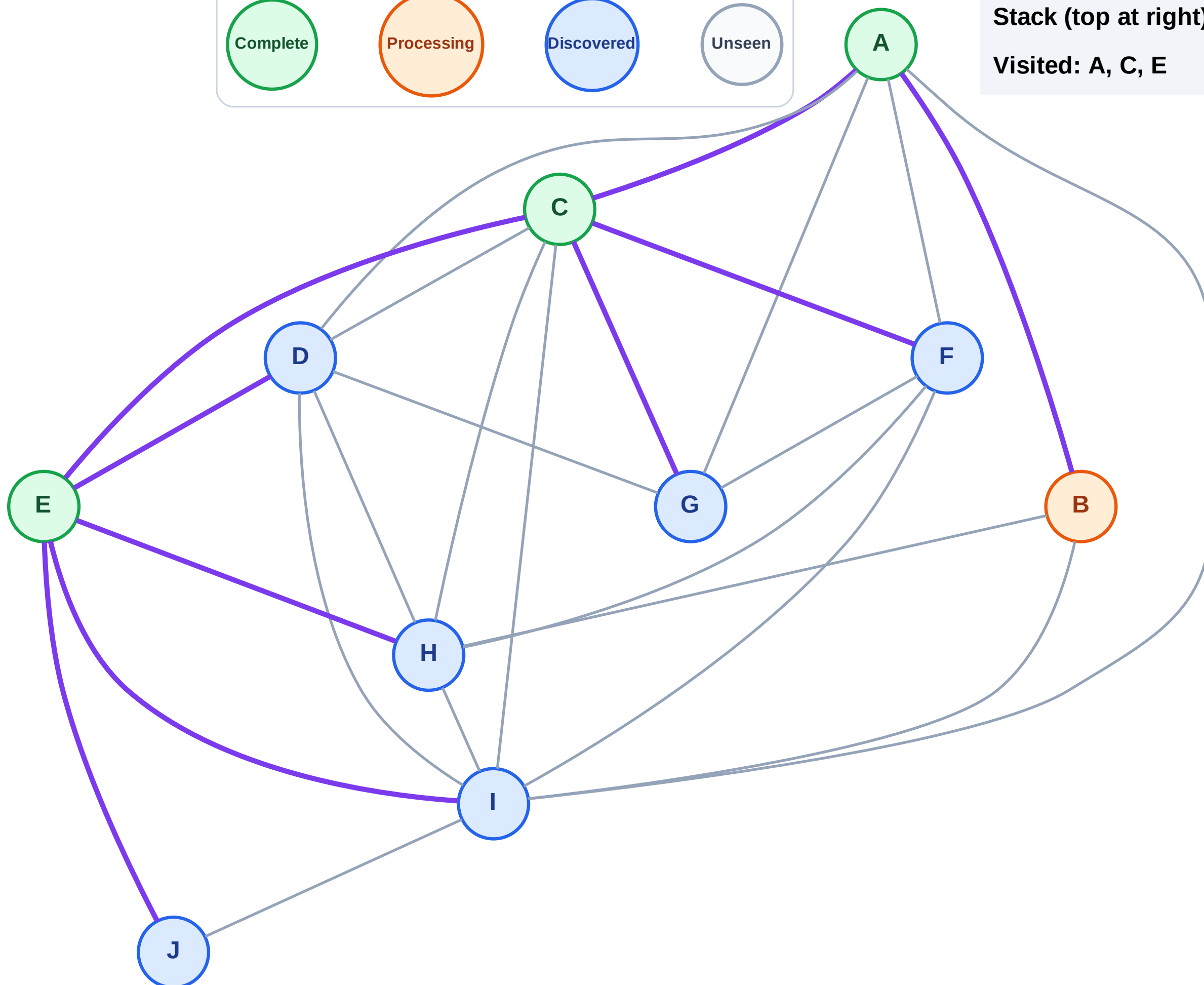
Discovered



Unseen

Stack (top at right): J | I | H | D | G | F

Visited: A, C, E



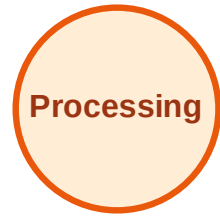
DFS Traversal

Step 9: No new neighbors from B

Legend



Complete



Processing



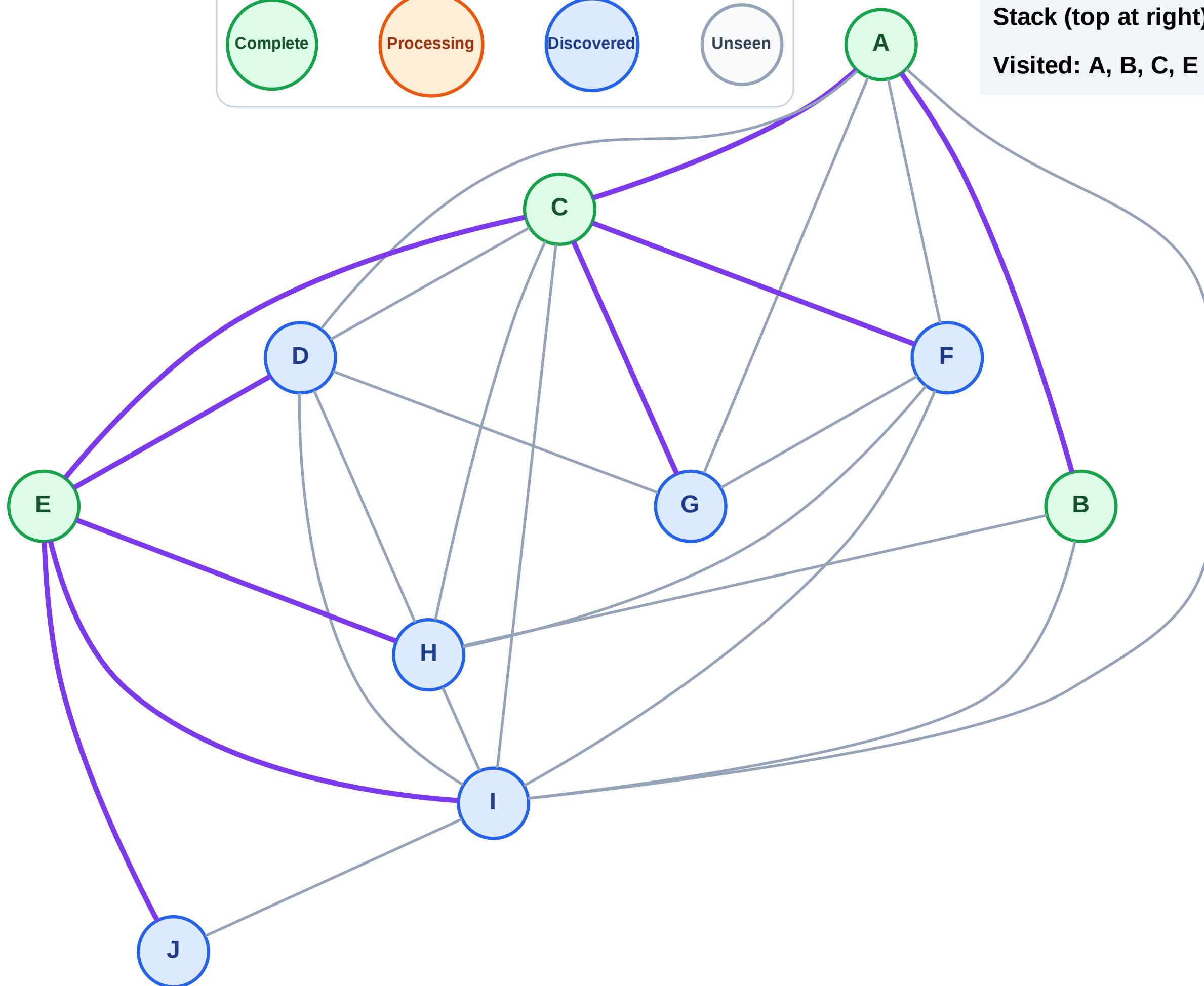
Discovered



Unseen

Stack (top at right): J | I | H | D | G | F

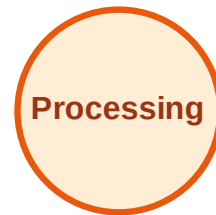
Visited: A, B, C, E



DFS Traversal

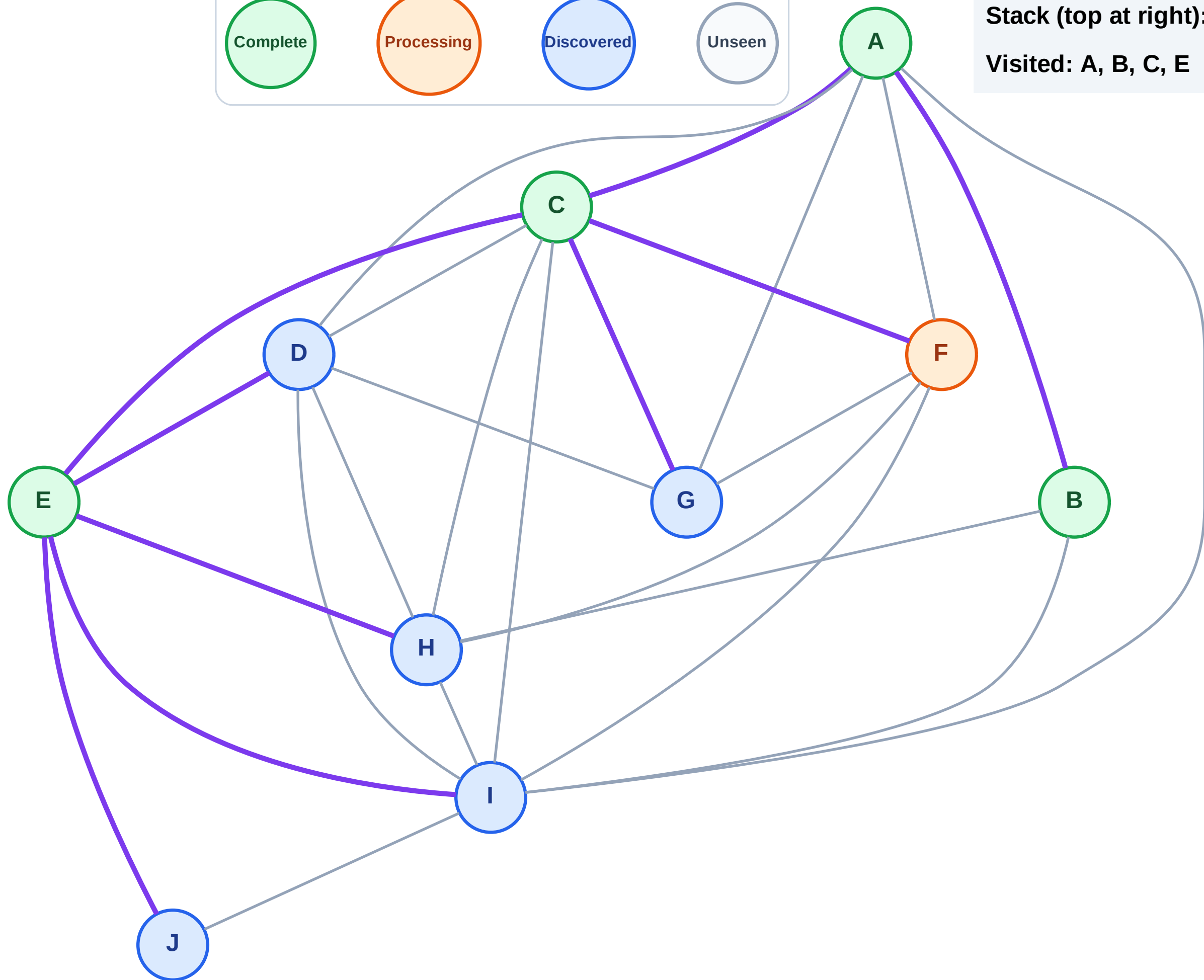
Step 10: Process node F

Legend



Stack (top at right): J | I | H | D | G

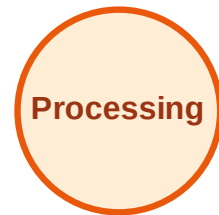
Visited: A, B, C, E



DFS Traversal

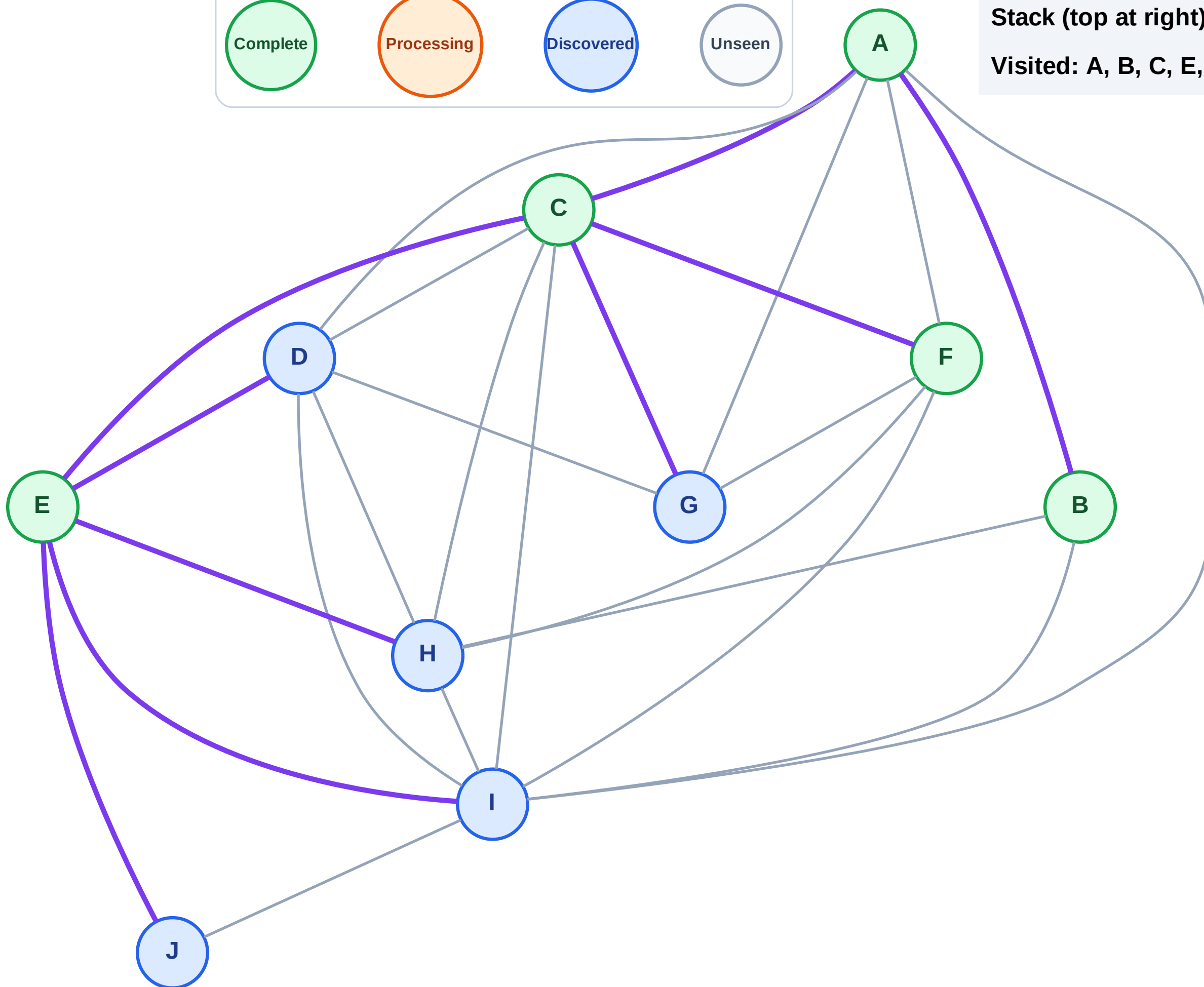
Step 11: No new neighbors from F

Legend



Stack (top at right): J | I | H | D | G

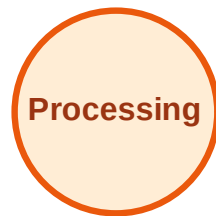
Visited: A, B, C, E, F



DFS Traversal

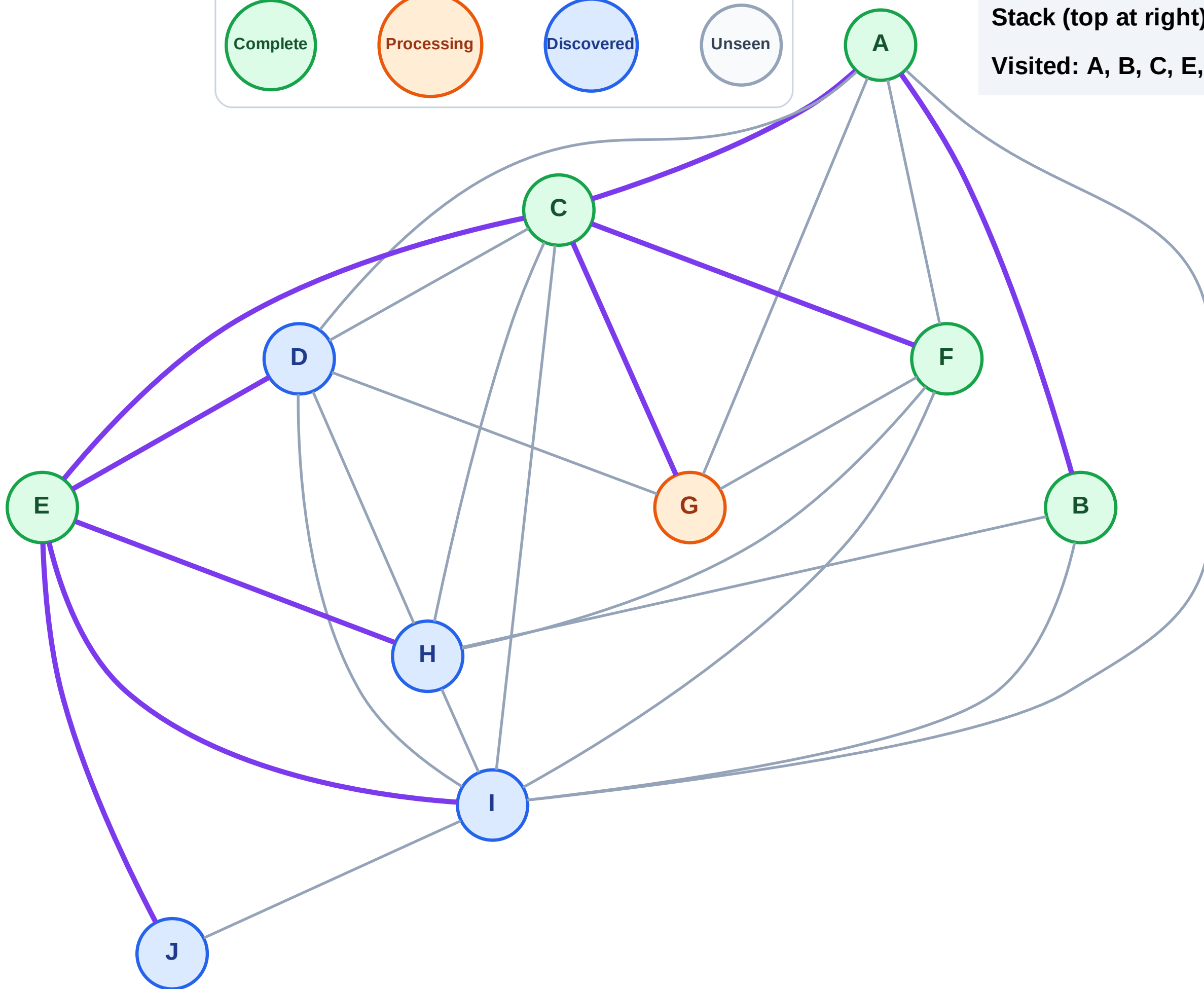
Step 12: Process node G

Legend



Stack (top at right): J | I | H | D

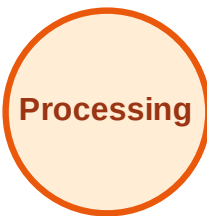
Visited: A, B, C, E, F



DFS Traversal

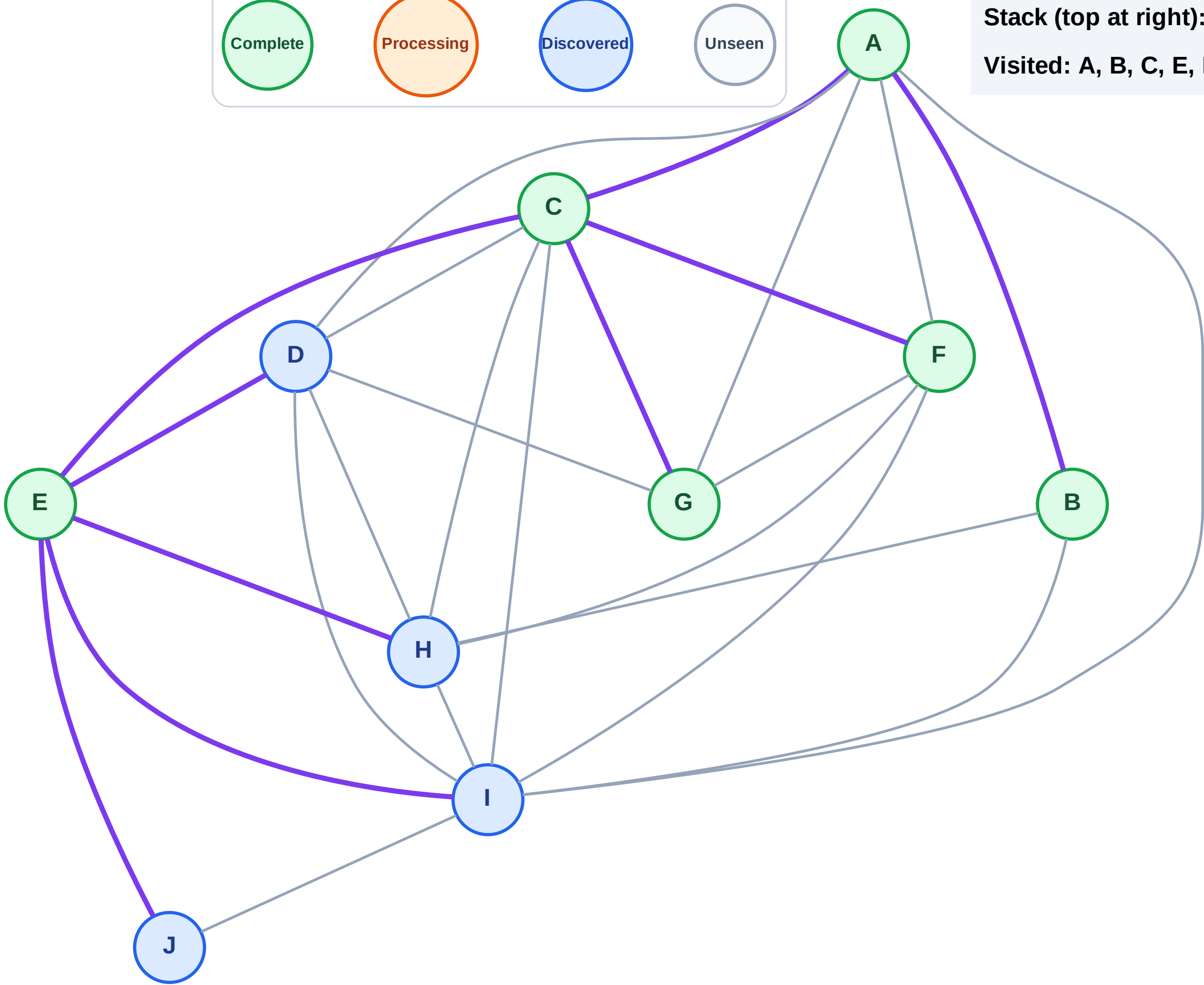
Step 13: No new neighbors from G

Legend



Stack (top at right): J | I | H | D

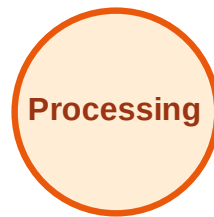
Visited: A, B, C, E, F, G



DFS Traversal

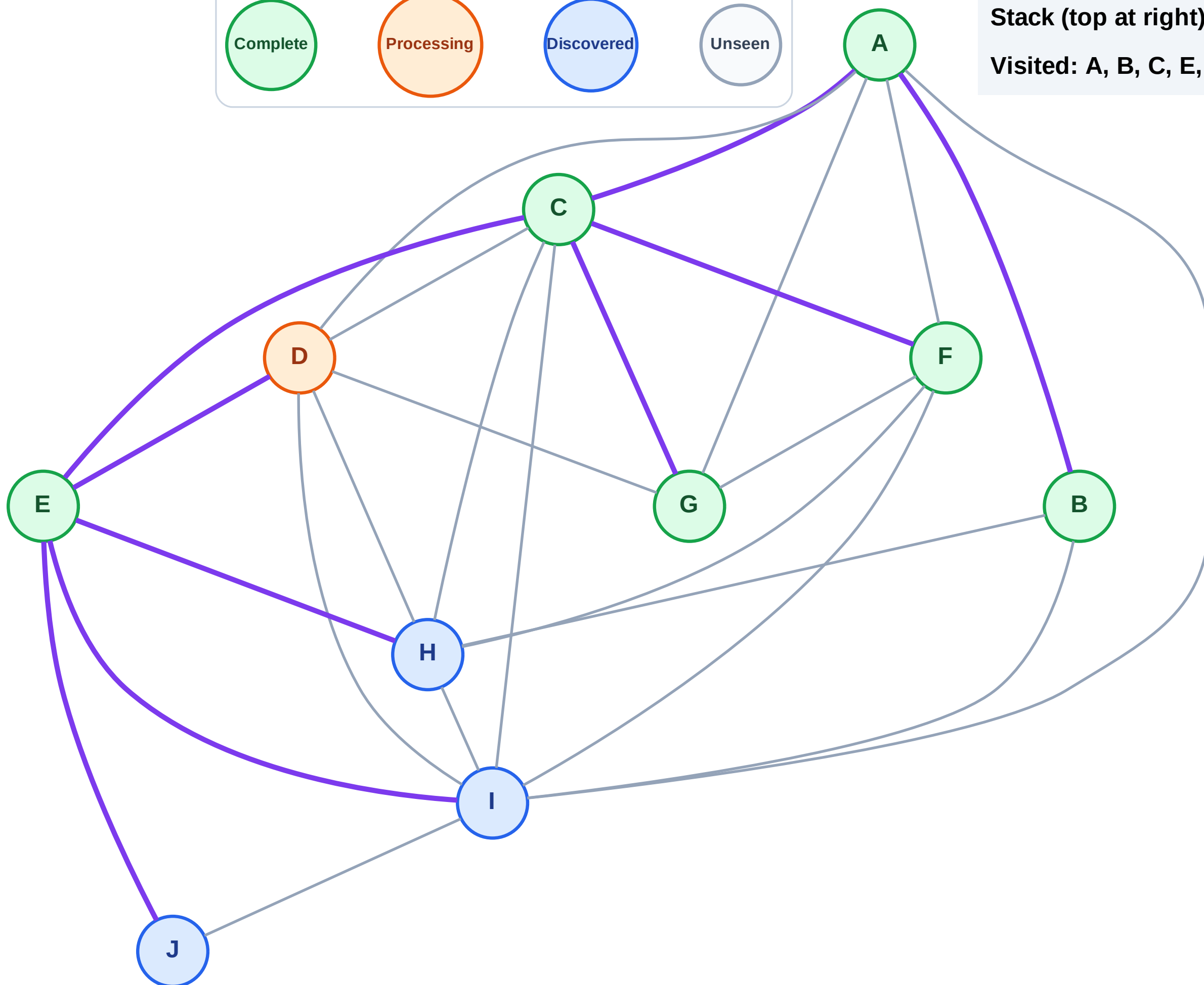
Step 14: Process node D

Legend



Stack (top at right): J | I | H

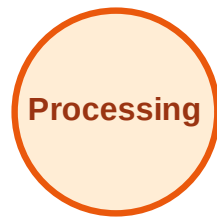
Visited: A, B, C, E, F, G



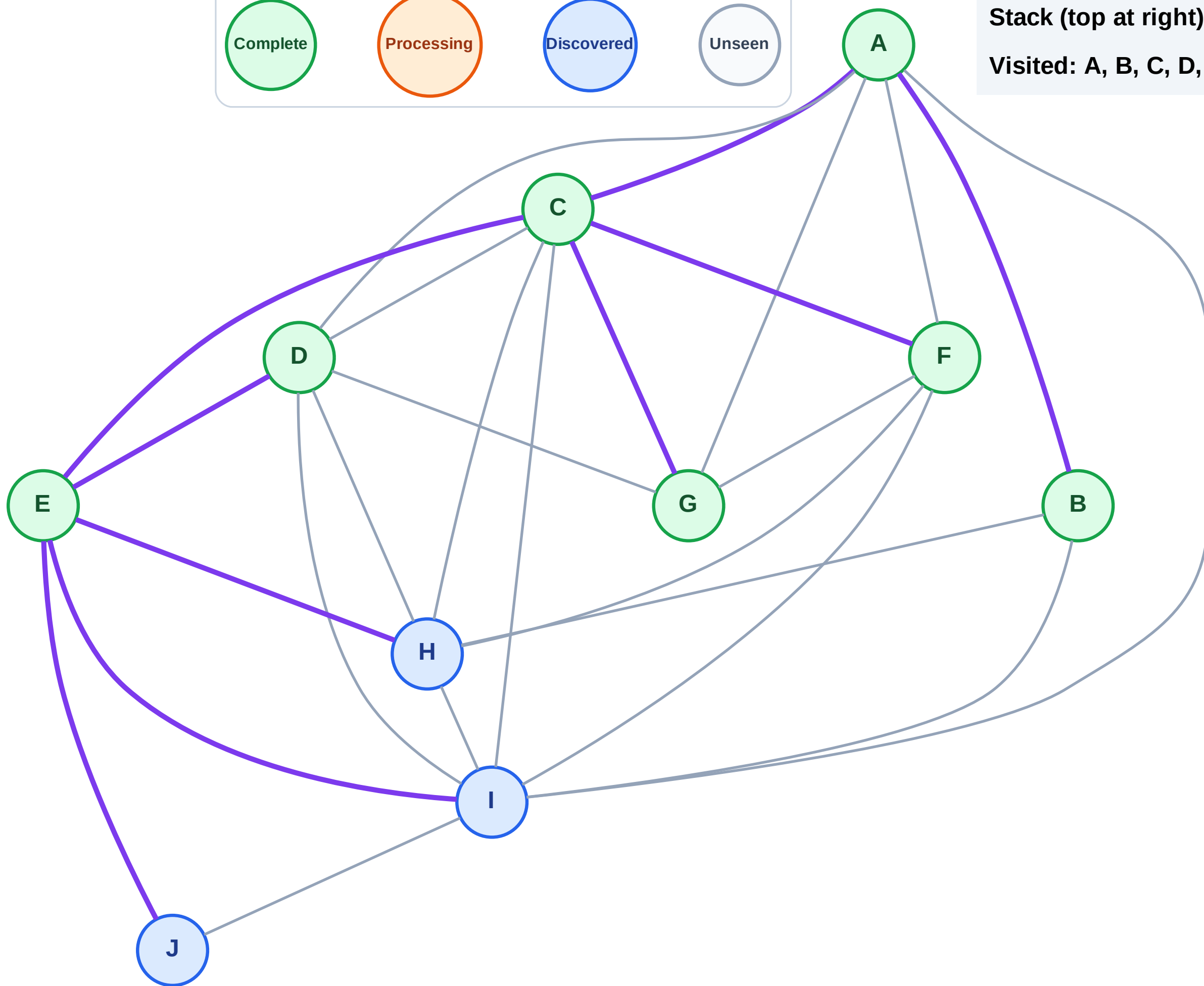
DFS Traversal

Step 15: No new neighbors from D

Legend



Stack (top at right): J | I | H
Visited: A, B, C, D, E, F, G



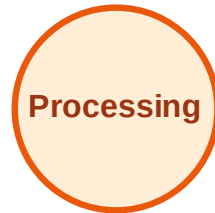
DFS Traversal

Step 16: Process node H

Legend



Complete



Processing

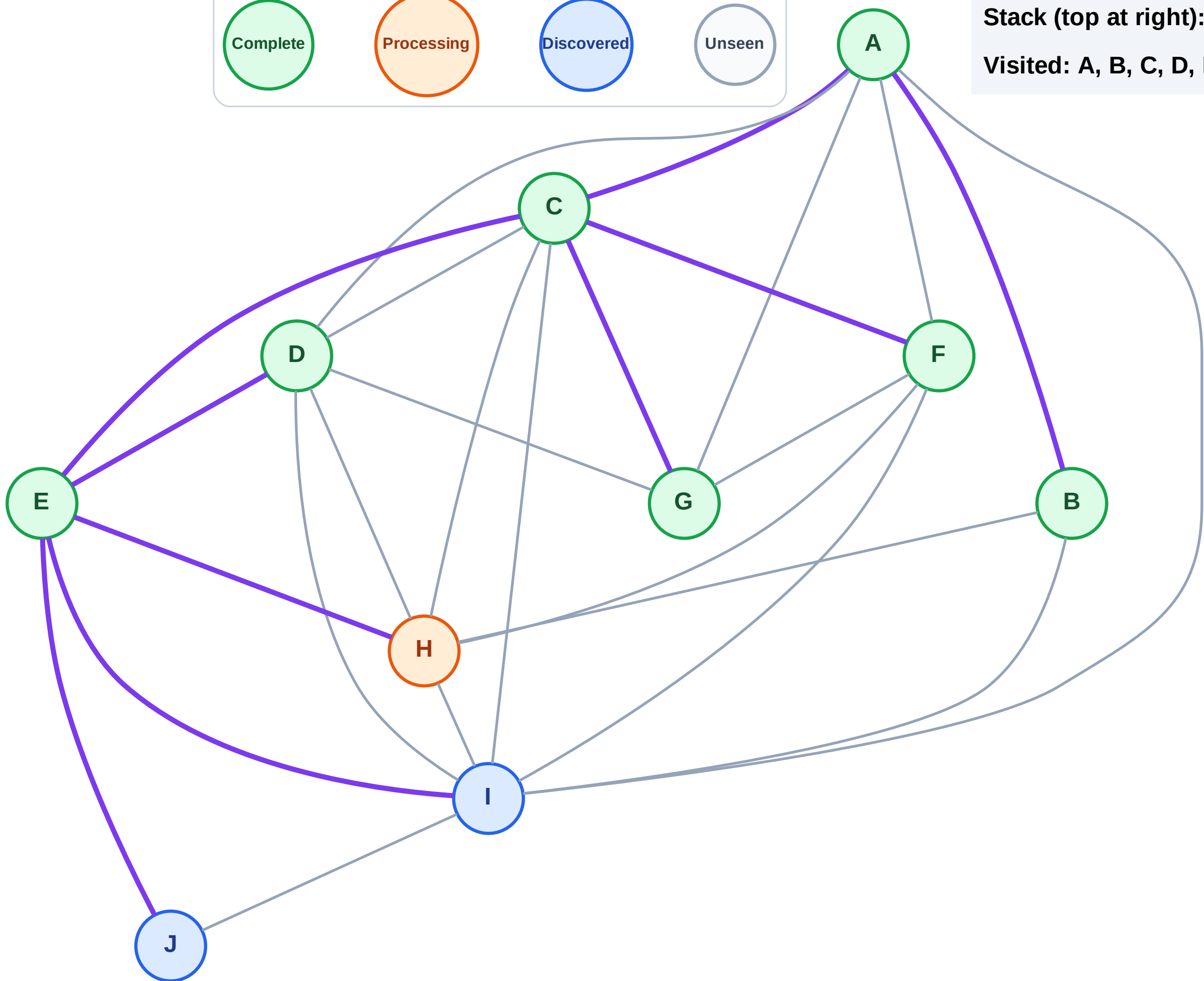


Discovered



Unseen

Stack (top at right): J | I
Visited: A, B, C, D, E, F, G



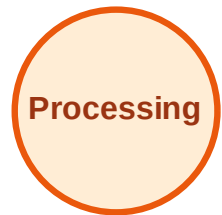
DFS Traversal

Step 17: No new neighbors from H

Legend



Complete



Processing

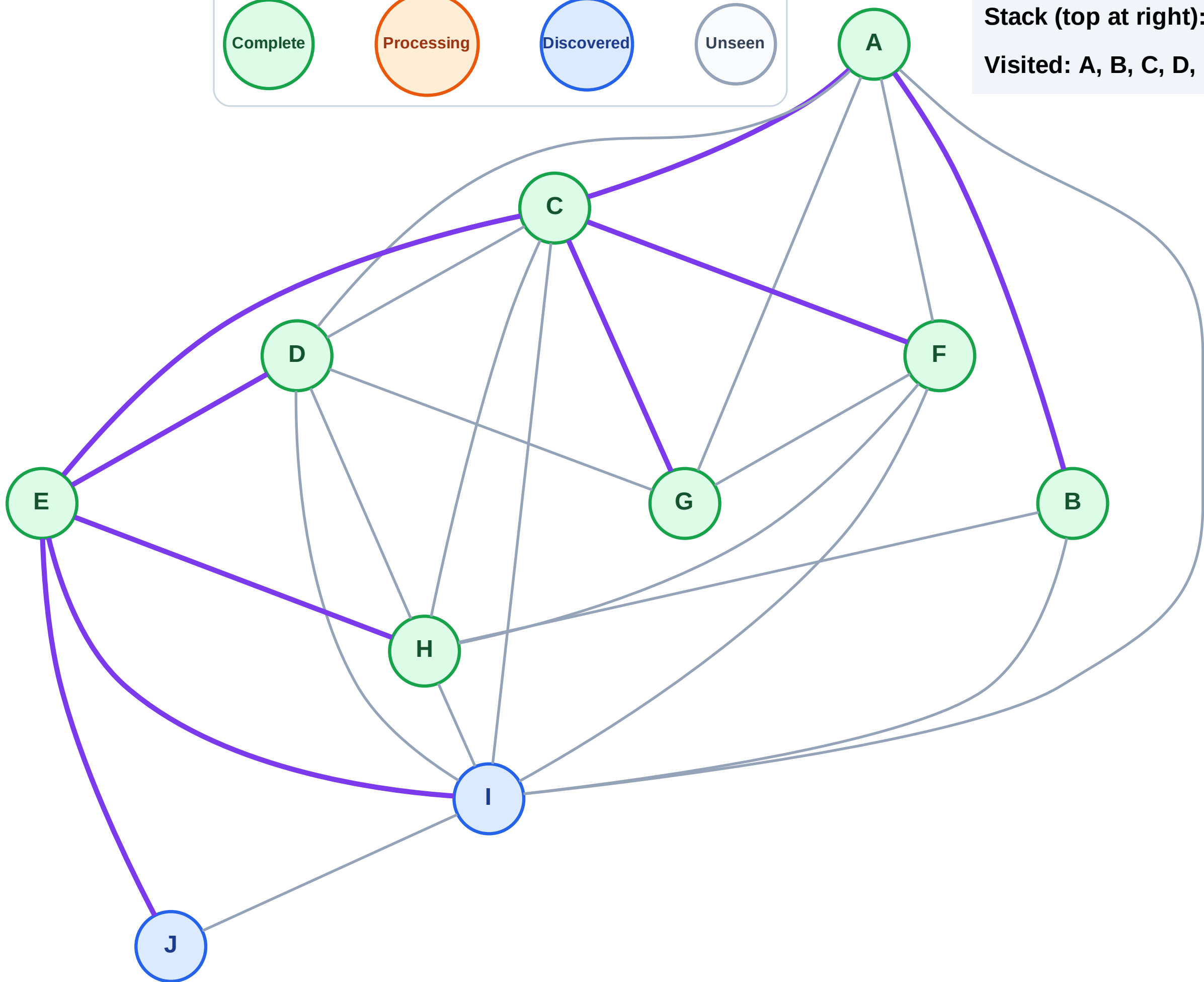


Discovered



Unseen

Stack (top at right): J | I
Visited: A, B, C, D, E, F, G, H



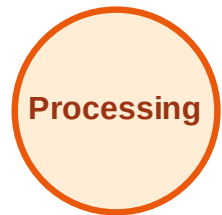
DFS Traversal

Step 18: Process node I

Legend



Complete



Processing



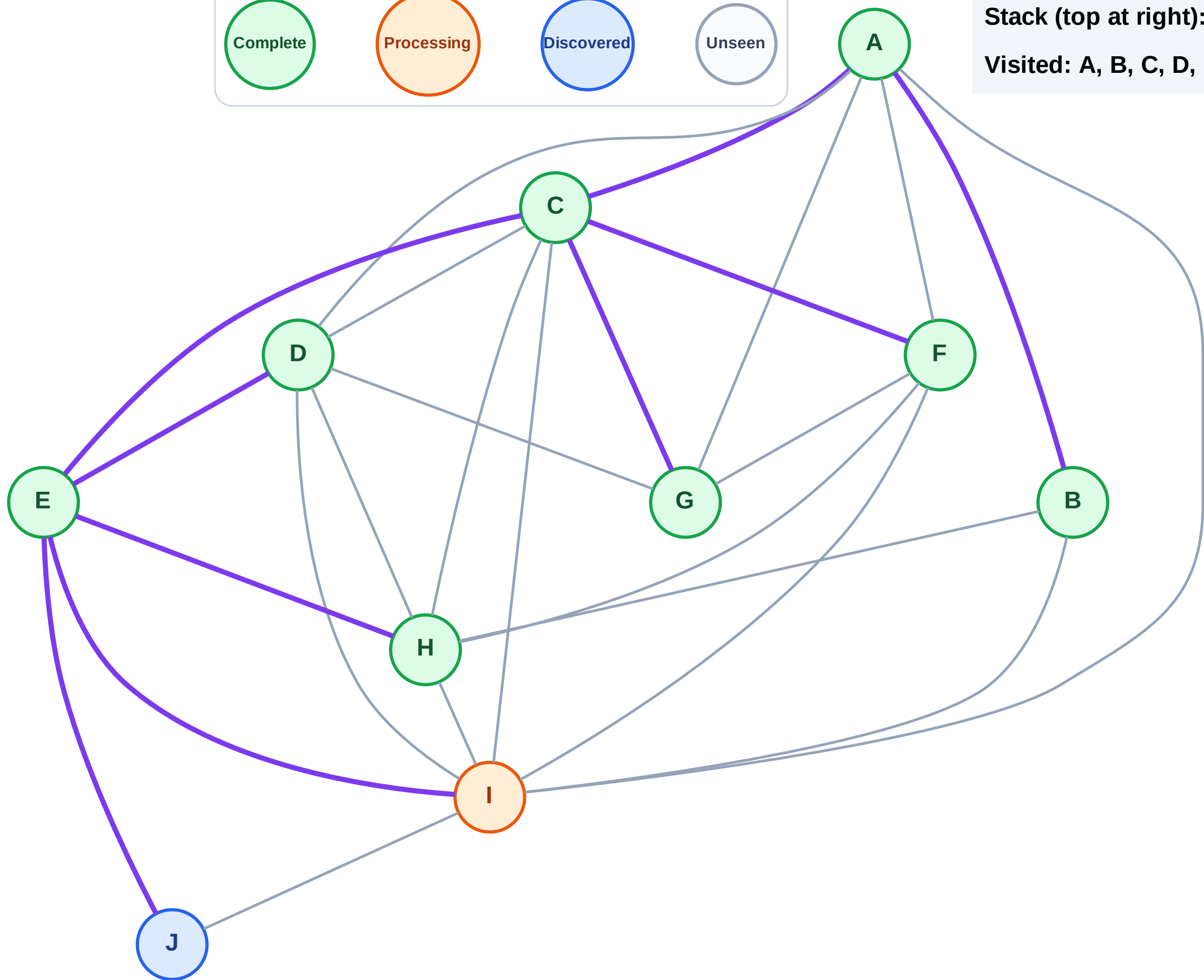
Discovered



Unseen

Stack (top at right): J

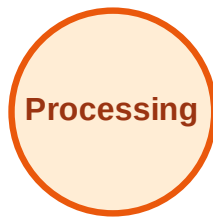
Visited: A, B, C, D, E, F, G, H



DFS Traversal

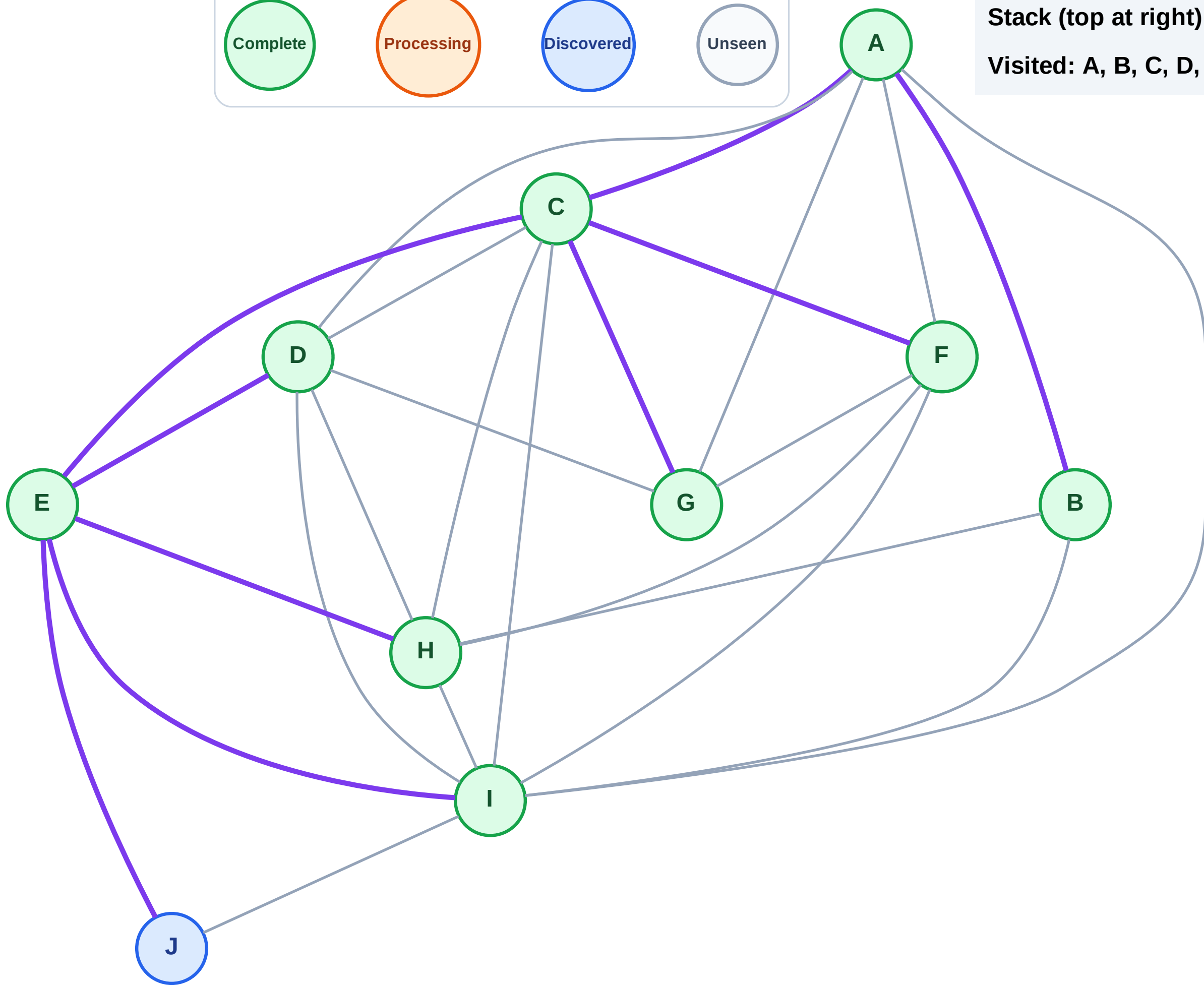
Step 19: No new neighbors from I

Legend



Stack (top at right): J

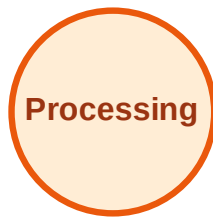
Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

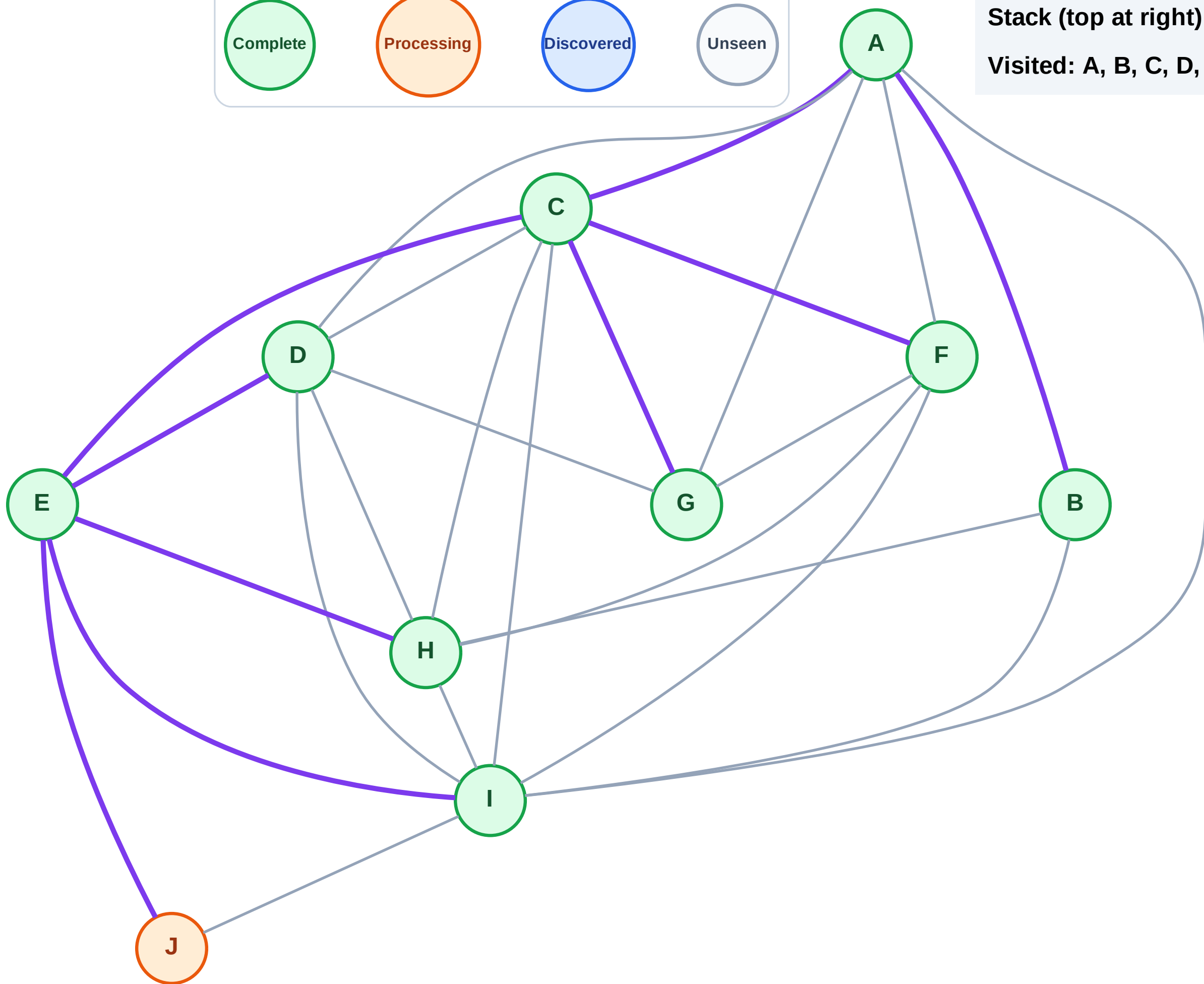
Step 20: Process node J

Legend



Stack (top at right): (empty)

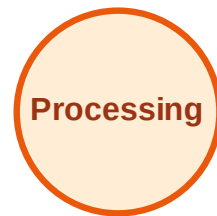
Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

Step 21: No new neighbors from J

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

